

Connectedness thresholds for pseudospectra of tridiagonal Toeplitz matrices

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Abstract

We study the open Euclidean-norm pseudospectra of $A_n(a) = \text{tridiag}(a, 0, 1)$, where $0 < a < 1$. For each order n , connectedness is equivalent to $\varepsilon > \gamma_n(a)$, where $\gamma_n(a)$ is the maximum of $s_{\min}(xI_n - A_n(a))$ over the real spectral interval. Every pseudospectral component is contractible. The thresholds decrease strictly with n , so connectedness, once attained, persists at every larger order. The proof uses vertical monotonicity of the least singular value, a symmetric matrix pencil, and separate comparisons for noncentral and central spectral gaps. Explicit upper and lower bounds show that $\gamma_n(a) \asymp_a (n+1)a^{n/2}$. Consequently, the first connected order equals $2[\log(1/\varepsilon) + \log \log(1/\varepsilon)] / \log(1/a) + O_a(1)$ as ε tends to zero. We also give the connectedness thresholds at $a = 0$ and $a = 1$.

Keywords: pseudospectrum; tridiagonal Toeplitz matrix; least singular value; connectedness

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1. Introduction

For a nonnormal matrix, small perturbations can move eigenvalues far from its spectrum. The pseudospectrum records this sensitivity, and its connected components describe which spectral clusters remain separated at a prescribed perturbation level. This paper determines when the open Euclidean-norm pseudospectrum of

$$A_n(a) = \text{tridiag}(a, 0, 1) = \begin{bmatrix} 0 & 1 & & & \\ a & 0 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & & \ddots & 1 \\ & & & a & 0 \end{bmatrix}, \quad n \geq 2, \quad 0 < a < 1, \quad (1.1)$$

is connected, and proves that connectedness persists as the order increases. The arguments of tridiag specify the subdiagonal, diagonal, and superdiagonal, respectively.

Throughout, $\|\cdot\|_2$ denotes the Euclidean vector norm or its induced matrix norm, $s_{\min}$ the least singular value, and I_n the identity matrix of order n . We use the open pseudospectrum

$$\sigma_\varepsilon(A_n) = \{z \in \mathbb{C} : s_{\min}(zI_n - A_n) < \varepsilon\}, \quad \varepsilon > 0. \quad (1.2)$$

It is also the union of $\sigma(A_n + E)$ over complex matrices E with $\|E\|_2 < \varepsilon$, or, equivalently, the spectrum together with the set where $\|(zI_n - A_n)^{-1}\|_2 > \varepsilon^{-1}$ [1]. Thus the perturbations in this paper are unrestricted complex matrices, measured in the induced Euclidean norm.

Toeplitz pseudospectra have long provided examples of the difference between spectral location and perturbation sensitivity. Resolvent estimates and pseudospectral behavior of finite non-Hermitian Toeplitz matrices are studied in [2]. Limiting results for pseudospectra and singular values of large

truncated Wiener–Hopf operators are developed in [3]. Explicit eigensystems and sensitivity formulas for tridiagonal Toeplitz matrices are given in [4]. Algorithms for the extreme points of Toeplitz-structured pseudospectra in the Frobenius norm are developed in [5]. Finite-section methods also provide convergent spectral and pseudospectral inclusion sets for banded and band-dominated operators [6].

The geometry of pseudospectra is closely connected with matrix nearness problems. Their component structure and variational properties are studied in [7]. For a matrix with distinct eigenvalues, the first coalescence level of pseudospectral components equals the distance to matrices with a multiple eigenvalue [8]. First-coalescence points are lowest generalized saddle points of the least-singular-value function [9]. Here the question is when all components have joined, and how that level changes between consecutive matrix orders.

Put $r = \sqrt{a}$, and let $V_n = \text{tridiag}(0, 0, 1)$. The diagonal matrix $\mathcal{G}_n(r) = \text{diag}(1, r, \dots, r^{n-1})$ gives

$$A_n = \mathcal{G}_n(r) r (V_n + V_n^T) \mathcal{G}_n(r)^{-1}. \quad (1.3)$$

The spectrum is therefore

$$\lambda_{n,k} = 2r \cos \frac{k\pi}{n+1}, \quad 1 \leq k \leq n. \quad (1.4)$$

Write

$$\begin{aligned} I_n^{\text{sp}} &= [\lambda_{n,n}, \lambda_{n,1}], & g_n(x) &= s_{\min}(xI_n - A_n), \\ \gamma_n(a) &= \max_{x \in I_n^{\text{sp}}} g_n(x), & N_c(a, \varepsilon) &= \min\{n \geq 2 : \sigma_\varepsilon(A_n(a)) \text{ is connected}\}. \end{aligned} \quad (1.5)$$

The maximum exists by continuity. The following theorem also ensures that N_c is finite. When a is fixed, we suppress it in A_n , g_n , and γ_n . We set $A_1 = V_1 = [0]$ when a smaller block occurs in a proof.

Theorem 1.1. *Let $0 < a < 1$. For every $n \geq 2$ and $\varepsilon > 0$, each connected component of $\sigma_\varepsilon(A_n(a))$ is contractible, and*

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff \varepsilon > \gamma_n(a). \quad (1.6)$$

Moreover,

$$\gamma_{n+1}(a) < \gamma_n(a), \quad \gamma_2(a) = a, \quad \lim_{n \rightarrow \infty} \gamma_n(a) = 0. \quad (1.7)$$

Consequently,

$$\{n \geq 2 : \sigma_\varepsilon(A_n(a)) \text{ is connected}\} = \{n \geq N_c(a, \varepsilon)\},$$

and $N_c(a, \varepsilon) = 2$ if and only if $\varepsilon > a$. For fixed a ,

$$N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log \log(1/\varepsilon)) + O_a(1) \quad (\varepsilon \downarrow 0). \quad (1.8)$$

The criterion (1.6) follows from contraction toward the real axis. The main comparison is $\gamma_{n+1} < \gamma_n$: both the spectral intervals and their gaps change with n , and the central gaps change with parity. We compare noncentral gaps through a parametrization of a symmetric matrix pencil, then treat the central gaps separately. Finally, explicit bounds $\gamma_n(a) \asymp_a (n+1)a^{n/2}$ give (1.8).

Section 2 proves contraction and the connectedness criterion. Section 3 establishes the strict comparison, including the continuant identities used in its proof. Section 4 gives quantitative bounds and proves the remaining assertions of the theorem. The boundary cases $a = 0, 1$ and numerical illustrations follow in Sections 5 and 6.

2. Contraction toward the real axis

The least singular value is nondecreasing along every vertical half-line issuing from the real axis. We prove this through a determinant identity and then use it to describe the pseudospectral components.

2.1. A determinant identity

Proposition 2.1 (Positivity of the determinant derivative). *Fix $0 < a < 1$, $x \in \mathbb{R}$, $Y \geq 0$, and $t \in \mathbb{R}$. For $k \geq 1$, set*

$$\begin{aligned}\mathcal{P}_k(Y, t) &= ((x + i\sqrt{Y})I_k - A_k)^*((x + i\sqrt{Y})I_k - A_k) - t^2 I_k, \\ F_k(Y, t) &= \det \mathcal{P}_k(Y, t),\end{aligned}$$

and set $F_0 = 1$. For every $n \geq 1$,

$$\mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0.$$

Proof. At a fixed value of Y , write

$$\delta_{\text{pent}} = x^2 + Y + 1 + a^2 - t^2, \quad b = -(1 + a)x + i(1 - a)\sqrt{Y}.$$

For $n \geq 2$, the matrix $\mathcal{P}_n(Y, t)$ has diagonal

$$\delta_{\text{pent}} - 1, \delta_{\text{pent}}, \dots, \delta_{\text{pent}}, \delta_{\text{pent}} - a^2,$$

whereas for $n = 1$ its sole entry is $\delta_{\text{pent}} - 1 - a^2$. In all dimensions its first upper diagonal is b , its first lower diagonal is $\bar{b}$, and its second off-diagonals are a . Let T_k be the determinant of the pure pentadiagonal Toeplitz matrix with diagonal δ_{pent} and these two off-diagonals. Set $T_{-1} = 0$, $T_0 = 1$, and put

$$D_k = T_k - T_{k-1}, \quad R_k = T_k - a^2 T_{k-1}. \quad (2.1)$$

Since a determinant is affine in each diagonal entry, replacing the first diagonal entry δ_{pent} by $\delta_{\text{pent}} - 1$ subtracts the cofactor T_{k-1} , whereas replacing the last one by $\delta_{\text{pent}} - a^2$ subtracts $a^2 T_{k-1}$. Thus, for $k < n$, D_k and R_k are, respectively, the leading and trailing principal minors of $\mathcal{P}_n(Y, t)$.

The following continuant calculation isolates all dependence on Y . With $T_k = 0$ for $k < 0$, a Laplace expansion over the last two rows, grouped according to the largest displaced index in the contributing permutation, gives for every $k \geq 1$

$$\begin{aligned}T_k &= \delta_{\text{pent}} T_{k-1} - (|b|^2 - a^2) T_{k-2} - (2a^2 \delta_{\text{pent}} - a(b^2 + \bar{b}^2)) T_{k-3} \\ &\quad - a^2 (|b|^2 - a^2) T_{k-4} + a^4 \delta_{\text{pent}} T_{k-5} - a^6 T_{k-6} - a^2 \mathbf{1}_{\{k=2\}}.\end{aligned} \quad (2.2)$$

For example, $T_1 = \delta_{\text{pent}}$ and $T_2 = \delta_{\text{pent}}^2 - |b|^2$; these also verify the only inhomogeneous coefficient in (2.2). Multiplying the recurrence by z^k and summing gives

$$\sum_{k \geq 0} T_k z^k = \frac{1 - a^2 z^2}{Q(z, Y)}, \quad (2.3)$$

where

$$\begin{aligned}Q(z, Y) &= 1 - \delta_{\text{pent}} z + (|b|^2 - a^2) z^2 + (2a^2 \delta_{\text{pent}} - a(b^2 + \bar{b}^2)) z^3 \\ &\quad + a^2 (|b|^2 - a^2) z^4 - a^4 \delta_{\text{pent}} z^5 + a^6 z^6.\end{aligned}$$

Substituting (2.3) into (2.1), with $\mathcal{D}(z, Y) = \sum_{k \geq 0} D_k z^k$, gives the first identity in

$$\mathcal{D}(z, Y) = \frac{(1 - z)(1 - a^2 z^2)}{Q(z, Y)}, \quad \sum_{k \geq 0} F_k(Y, t) z^k = (1 - a^2 z) \mathcal{D}(z, Y). \quad (2.4)$$

For the second identity in (2.4), first change the initial diagonal, producing D_k , and then change the final diagonal; its cofactor is D_{k-1} . Thus $F_k = D_k - a^2 D_{k-1}$, which proves the asserted generating function. Direct substitution of δ_{pent} and b gives

$$\partial_Y Q = -z(1 - z)(1 + az)^2(1 - a^2 z).$$

Differentiating (2.4) therefore yields

$$\sum_{k \geq 0} \partial_Y F_k(Y, t) z^k = \varphi_a(z) \mathcal{D}(z, Y)^2, \quad \varphi_a(z) = \frac{z(1+az)(1-a^2z)^2}{1-az}. \quad (2.5)$$

Suppose that $\mathcal{P}_n(Y, t) > 0$. Then

$$D_k > 0, \quad R_k > 0 \quad (0 \leq k < n). \quad (2.6)$$

Under the positivity in (2.6), put $u_k = \sum_{j=0}^k D_j D_{k-j}$, with $u_k = 0$ for $k < 0$. To extract coefficients from (2.5), note that the coefficients of φ_a , beginning with the coefficient of z , are

$$1, \quad 2a(1-a), \quad a^2(a^2-4a+2), \quad 2a^{k-1}(1-a)^2 \quad (k \geq 4).$$

Thus

$$\begin{aligned} \partial_Y F_n &= u_{n-1} + 2a(1-a)u_{n-2} + a^2(a^2-4a+2)u_{n-3} \\ &\quad + 2(1-a)^2 \sum_{k=4}^n a^{k-1} u_{n-k}. \end{aligned} \quad (2.7)$$

The third coefficient is the sole term that may be negative; the following cofactor inequality controls its contribution. Let $j, k \geq 1$, $j+k = n-1$. Deleting index $k+1$ from $\mathcal{P}_n$ leaves blocks of sizes k and j , with the second off-diagonal a as their sole coupling. In block form, the deleted principal submatrix is

$$\begin{bmatrix} L_k & a\mathbf{e}_k\mathbf{e}_1^T \\ a\mathbf{e}_1\mathbf{e}_k^T & R_j^{\text{mat}} \end{bmatrix}, \quad \det L_k = D_k, \quad \det R_j^{\text{mat}} = R_j.$$

Expanding once in each of the two coupling entries shows that its determinant is

$$D_k R_j - a^2 D_{k-1} R_{j-1} > 0. \quad (2.8)$$

Deleting index $j+1$ instead gives the same inequality with j, k interchanged. Define

$$\vartheta_k = \frac{D_k}{D_{k-1}}, \quad \omega_k = \frac{R_k}{R_{k-1}}.$$

Since $T_k = \sum_{\ell=0}^k D_\ell$, (2.1) gives

$$R_k = D_k + (1-a^2) \sum_{\ell=0}^{k-1} D_\ell, \quad R_k - R_{k-1} = D_k - a^2 D_{k-1}. \quad (2.9)$$

The second identity in (2.9) shows that $\vartheta_k \leq a^2$ implies $\omega_k \leq 1$. On dividing (2.8) and its companion by positive factors, we obtain

$$\vartheta_k \omega_j > a^2, \quad \vartheta_j \omega_k > a^2. \quad (2.10)$$

If both ϑ_j and ϑ_k were at most a^2 , then both ω_j and ω_k would be at most 1, contradicting (2.10). Hence, after interchanging j, k if necessary, $\vartheta_j > a^2$. Moreover,

$$\vartheta_j R_{j-1} - a^2 R_j = (\vartheta_j - a^2)(R_{j-1} - a^2 D_{j-1}),$$

and the second factor equals $(1-a^2) \sum_{\ell=0}^{j-1} D_\ell > 0$. Therefore

$$\vartheta_j > a^2 \implies \omega_j < \vartheta_j / a^2.$$

The first inequality in (2.10) now gives $a^2 < \vartheta_k \omega_j < \vartheta_j \vartheta_k / a^2$, and therefore

$$\vartheta_j \vartheta_k > a^4. \quad (2.11)$$

Consequently, for $n \geq 3$, applying (2.11) to $j = p + 1, k = n - 2 - p$ gives

$$D_{p+1} D_{n-2-p} > a^4 D_p D_{n-3-p}, \quad 0 \leq p \leq n - 3.$$

Summing and then adding the two positive endpoint terms gives

$$u_{n-1} = 2D_{n-1} + \sum_{j=1}^{n-2} D_j D_{n-1-j} > a^4 u_{n-3}. \quad (2.12)$$

If the third coefficient in (2.7) is negative, write it as $-\mathcal{B}_{\text{neg}}$, where $\mathcal{B}_{\text{neg}} = a^2(4a - 2 - a^2)$. Since

$$a^4 - \mathcal{B}_{\text{neg}} = 2a^2(1 - a)^2 > 0,$$

equations (2.7) and (2.12) prove (2.13). If that coefficient is nonnegative, the implication follows directly. For $n = 1$, one has $\partial_Y F_1 = 1$; for $n = 2$, one has $\partial_Y F_2 = 2D_1 + 2a(1 - a) > 0$. Thus (2.13) holds in every dimension, which proves the proposition. $\square$

2.2. Vertical monotonicity

Lemma 2.2 (Vertical monotonicity). *Let $n \geq 2$ and $0 < a < 1$. For every real x ,*

$$y \longmapsto s_{\min}((x + iy)I_n - A_n), \quad y \geq 0,$$

is nondecreasing.

Proof. Put $Y = y^2$, and, for $t \geq 0$, define

$$\begin{aligned} \mathcal{P}_n(Y, t) &= ((x + i\sqrt{Y})I_n - A_n)^*((x + i\sqrt{Y})I_n - A_n) - t^2 I_n, \\ F_n(Y, t) &= \det \mathcal{P}_n(Y, t). \end{aligned}$$

Proposition 2.1 proves the finite-dimensional implication

$$\mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0. \quad (2.13)$$

Fix $Y_0 \geq 0$. When the least singular value at Y_0 equals zero, monotonicity follows from nonnegativity of singular values. Otherwise take $0 \leq t < s_{\min}((x + i\sqrt{Y_0})I_n - A_n)$. On the maximal interval to the right of Y_0 on which $\mathcal{P}_n(Y, t) > 0$, implication (2.13) makes its determinant strictly increasing. At any finite right endpoint, continuity of Hermitian eigenvalues would produce a positive-semidefinite singular matrix with determinant zero. Strict increase from the positive determinant at Y_0 precludes such an endpoint. Thus $\mathcal{P}_n(Y, t) > 0$ for every $Y \geq Y_0$. Letting t increase to the least singular value at Y_0 proves the lemma. $\square$

2.3. Components and the real-axis criterion

We first recall the component property of finite-dimensional pseudospectra [7] and include its short proof.

Lemma 2.3. *Every connected component of $\sigma_\epsilon(A_n)$ contains an eigenvalue of A_n .*

Proof. The estimate $s_{\min}(zI_n - A_n) \geq |z| - \|A_n\|_2$ makes the pseudospectrum bounded. Suppose that a component $\mathcal{C}$ contains no eigenvalue. Its closure contains no eigenvalue either: a sufficiently small

pseudospectral disk about such an eigenvalue would meet $\mathcal{C}$ and hence belong to the same component. Thus the resolvent is analytic on a neighborhood of $\overline{\mathcal{C}}$. At every boundary point its norm is at most ε^{-1} . Indeed, a boundary point in the open pseudospectrum would have a small connected neighborhood in that set and would belong to $\mathcal{C}$. The resolvent norm is subharmonic, so the maximum principle bounds it by ε^{-1} throughout $\mathcal{C}$, a contradiction. $\square$

Proposition 2.4. *Each component of $\sigma_\varepsilon(A_n)$ is contractible, and (1.6) holds.*

Proof. Since A_n is real, conjugation preserves the least singular value. Lemma 2.2 therefore shows that

$$H_t(x + iy) = x + i(1 - t)y, \quad 0 \leq t \leq 1,$$

maps $\Omega = \sigma_\varepsilon(A_n)$ into itself. For any component $\mathcal{C}$, the path $t \mapsto H_t(z)$ remains in $\mathcal{C}$. The restriction of H_t is a strong deformation retraction onto $\mathcal{C} \cap \mathbb{R}$. This intersection is nonempty and connected, being the image of $\mathcal{C}$ under H_1 ; hence it is an interval. It follows that $\mathcal{C}$ is contractible.

If Ω is connected, its projection $\Omega \cap \mathbb{R}$ is an interval containing all eigenvalues, and therefore contains I_n^{sp} . Conversely, if $I_n^{\text{sp}} \subset \Omega$, that interval belongs to one component and contains every eigenvalue. Lemma 2.3 excludes any other component. Thus connectedness is equivalent to $g_n(x) < \varepsilon$ on the compact interval I_n^{sp} , or $\gamma_n < \varepsilon$. At equality a maximizing point is excluded, so the open pseudospectrum is disconnected. $\square$

3. Strict matrix-order comparison of real-gap barriers

The global barrier may occur in any inter-eigenvalue gap, while the central gap changes with parity. We therefore compare every gap. Persymmetry selects the signed eigenvalue branch whose magnitude is the least singular value; folded Chebyshev coordinates and chord geometry compare the noncentral gaps, and a separate argument treats the parity-changing central gaps. Write the eigenvalues in decreasing order as

$$\lambda_{n,k} = 2r \cos \frac{k\pi}{n+1}, \quad 1 \leq k \leq n,$$

and call

$$G_{n,k} = [\lambda_{n,k+1}, \lambda_{n,k}], \quad \gamma_{n,k} = \max_{x \in G_{n,k}} g_n(x), \quad 1 \leq k < n,$$

the closed spectral gaps and their heights. Continuity guarantees that each maximum is attained. The spectral endpoints have height zero, whereas $g_n(x) > 0$ in every open gap. Thus every gap height is positive, every maximizer lies in the open gap, and

$$\gamma_n = \max_{1 \leq k < n} \gamma_{n,k}. \quad (3.1)$$

Lemma 3.1 (Strict barrier comparison). *For every $n \geq 2$,*

$$\gamma_{n+1} < \gamma_n.$$

3.1. Signed-pencil representation and branch selection

Let J_n be the reversal permutation matrix, so that $J_n \mathbf{e}_j = \mathbf{e}_{n+1-j}$. Persymmetry supplies the signed spectral representation needed for the matrix-order comparison: $A_n J_n = J_n A_n^T$, so $B_n(x) = (xI_n - A_n)J_n$ is real symmetric. Moreover,

$$(xI_n - A_n - sJ_n)J_n = B_n(x) - sI_n.$$

Since J_n is orthogonal, the absolute values of the eigenvalues of $B_n(x)$ are exactly the singular values of $xI_n - A_n$, including multiplicity. Consequently,

$$s_{\min}(xI_n - A_n) = \min\{|s| : \det(xI_n - A_n - sJ_n) = 0\}. \quad (3.2)$$

Set $q_n(x, s) = \det(xI_n - A_n - sJ_n)$; the minimum in (3.2) selects its signed branch nearest zero.

Lemma 3.2 (The middle signed branch). *Write the eigenvalues of $B_n(x)$ in decreasing order as $\mu_1(x) \geq \dots \geq \mu_n(x)$, and abbreviate $\lambda_k = \lambda_{n,k}$. If $\lambda_{j+1} < x < \lambda_j$, then*

$$s_{\min}(xI_n - A_n) = |s_*(x)|, \quad \text{sgn}(s_*(x)) = (-1)^j,$$

where

$$s_*(x) = \begin{cases} \mu_m(x), & n = 2m, \\ \mu_{m+1}(x), & n = 2m + 1. \end{cases}$$

Proof. Let $A_n^{\text{sym}} = r(V_n + V_n^T)$. The identities

$$\begin{aligned} A_n &= \mathcal{G}_n(r) A_n^{\text{sym}} \mathcal{G}_n(r)^{-1}, \\ \mathcal{G}_n(r)^{-1} J_n &= r^{-(n-1)} J_n \mathcal{G}_n(r) \end{aligned}$$

give

$$B_n(x) = r^{-(n-1)} \mathcal{G}_n(r) (xI_n - A_n^{\text{sym}}) J_n \mathcal{G}_n(r). \quad (3.3)$$

The sine eigenvector of A_n^{sym} belonging to λ_k is an eigenvector of J_n with eigenvalue $(-1)^{k+1}$. Hence (3.3) and Sylvester's law of inertia show that $B_n(x)$ has the same inertia as the diagonal matrix with entries

$$(-1)^{k+1}(x - \lambda_k), \quad 1 \leq k \leq n.$$

For a real symmetric matrix H , let $v_+(H)$ denote the number of its positive eigenvalues, counted with multiplicity. Counting signs gives

$$v_+(B_{2m}(x)) = \begin{cases} m, & j \text{ even}, \\ m-1, & j \text{ odd}, \end{cases} \quad v_+(B_{2m+1}(x)) = \begin{cases} m+1, & j \text{ even}, \\ m, & j \text{ odd}. \end{cases} \quad (3.4)$$

First let $n = 2m$. Permute the coordinates into the two groups

$$(1, \dots, m), \quad (2m, 2m-1, \dots, m+1).$$

Here $\mathbf{e}_j$ denotes the j th Euclidean coordinate vector in the ambient space. Put $P_m(x) = xI_m - A_m$. In these coordinates

$$\begin{aligned} B_{2m}(x) &= \hat{P}_m(x) - \mathcal{R}_{\text{bd}}, \\ \hat{P}_m(x) &= \begin{bmatrix} 0 & P_m(x) \\ P_m(x)^T & 0 \end{bmatrix}, \\ \mathcal{R}_{\text{bd}} &= \begin{bmatrix} \mathbf{e}_m \mathbf{e}_m^T & 0 \\ 0 & a \mathbf{e}_m \mathbf{e}_m^T \end{bmatrix}, \end{aligned} \quad (3.5)$$

In (3.5), $\mathcal{R}_{\text{bd}} \geq 0$ and $\text{rank } \mathcal{R}_{\text{bd}} = 2$. For $S = \text{diag}(I_m, -I_m)$,

$$\tilde{B} = -SB_{2m}(x)S = \hat{P}_m(x) + \mathcal{R}_{\text{bd}} = B_{2m}(x) + 2\mathcal{R}_{\text{bd}}.$$

If $\tilde{\mu}_1 \geq \cdots \geq \tilde{\mu}_{2m}$ are the eigenvalues of $\tilde{B}$, then

$$\tilde{\mu}_k = -\mu_{2m+1-k}. \quad (3.6)$$

Combining the reflection (3.6) with positive-semidefinite rank-two interlacing gives

$$\tilde{\mu}_k \geq \mu_k \geq \tilde{\mu}_{k+2}, \quad 1 \leq k \leq 2m-2. \quad (3.7)$$

If j is even, the only candidates for minimum modulus have moduli μ_m and $-\mu_{m+1} = \tilde{\mu}_m$, and $\mu_m \leq \tilde{\mu}_m$. If j is odd and $m \geq 2$, the two candidates have moduli μ_{m-1} and $-\mu_m = \tilde{\mu}_{m+1}$; taking $k = m-1$ in (3.7) gives $\mu_{m-1} \geq \tilde{\mu}_{m+1}$. For $m = 1$, both eigenvalues in the sole gap are negative, so the larger one μ_1 has minimum modulus. Thus μ_m always attains the minimum modulus, and its sign follows from (3.4). Equality in an interlacing comparison corresponds to a multiple least singular value.

Now let $n = 2m + 1$. Group the first and last m coordinates as above and leave the middle coordinate last. Then

$$B_{2m+1}(x) = \begin{bmatrix} \hat{P}_m(x) & w \\ w^T & x \end{bmatrix}, \quad w = \begin{bmatrix} -\mathbf{e}_m \\ -a\mathbf{e}_m \end{bmatrix}. \quad (3.8)$$

In an open spectral gap of A_{2m+1} , $P_m(x)$, and hence the principal block $\hat{P}_m(x)$ in (3.8), is nonsingular. Indeed, the explicit path spectrum gives

$$\sigma(A_m) = \left\{ 2r \cos \frac{\ell\pi}{m+1} : 1 \leq \ell \leq m \right\} = \{ \lambda_{2m+1,2\ell} : 1 \leq \ell \leq m \},$$

so its zeros are exactly the even-indexed eigenvalues of A_{2m+1} , none of which lies in an open spectral gap. If $s_1(P_m) \geq \cdots \geq s_m(P_m)$ are the singular values of $P_m(x)$, then the eigenvalues of the principal submatrix $\hat{P}_m(x)$ are

$$s_1(P_m) \geq \cdots \geq s_m(P_m) > 0 > -s_m(P_m) \geq \cdots \geq -s_1(P_m).$$

Cauchy interlacing gives

$$\mu_m \geq s_m(P_m) \geq \mu_{m+1} \geq -s_m(P_m) \geq \mu_{m+2}.$$

If j is even, (3.4) and interlacing give

$$0 < \mu_{m+1} \leq s_m(P_m) \leq -\mu_{m+2};$$

if j is odd, they give

$$0 < -\mu_{m+1} \leq s_m(P_m) \leq \mu_m.$$

Every other eigenvalue is farther from zero. Hence μ_{m+1} attains the minimum modulus, with sign prescribed by (3.4); equality corresponds to multiplicity. At a spectral endpoint the zero is simple because A_n is similar to the Jacobi matrix A_n^{sym} , whose eigenvalues are simple. Thus the selected ordered branch extends continuously to zero at both ends of the closed gap. This proves the assertion. $\square$

3.2. Folded continuants

We use folded coordinates for the noncentral gaps and a separate interlacing argument for the parity-changing central gaps. Let U_j be the Chebyshev polynomial of the second kind, with $U_{-1} = 0$. Put

$$C_0 = x^2 + (1-a)^2 - s^2, \quad \zeta_s = \frac{1+a^2+(1+a)s}{2a},$$

and define the folded discriminant and recurrence coefficient by

$$\Delta_{\text{fold}} = ((1+a-x)^2 - s^2)((1+a+x)^2 - s^2),$$

$$\mathfrak{b} = (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3.$$

The global recurrence polynomial is

$$\begin{aligned} \Pi(z) &:= 1 - C_0 z + \mathfrak{b} z^2 - a^2 C_0 z^3 + a^4 z^4, \\ u, v \in \mathbb{R}, \quad 2a(u+v) &= C_0, \quad a^2(4uv+2) = \mathfrak{b}, \\ \implies \Pi(z) &= (1 - 2auz + a^2 z^2)(1 - 2avz + a^2 z^2). \end{aligned} \quad (3.9)$$

On a real sheet with $\Delta_{\text{fold}} \geq 0$, we additionally put

$$\zeta_{\pm} = \frac{C_0 \pm \sqrt{\Delta_{\text{fold}}}}{4a}.$$

Then the pair $(u, v) = (\zeta_+, \zeta_-)$ satisfies the two symmetric relations in (3.9):

$$\zeta_+ + \zeta_- = \frac{C_0}{2a}, \quad \mathfrak{b} = a^2(4\zeta_+\zeta_- + 2).$$

We use the root presentation on these real sheets and the polynomial Π itself for the global transfer and generating identities. We will obtain the following formulas from a finite-dimensional transfer recurrence. For a differentiable real function f , write

$$[f]_{u,v} := \begin{cases} \frac{f(u) - f(v)}{u - v}, & u \neq v, \\ f'(u), & u = v \end{cases}.$$

Whenever the real pair u, v satisfies the two symmetric relations in (3.9), the folded determinants are

$$q_{2m}(x, s) = a^m [\mathfrak{f}_m]_{u,v}, \quad \mathfrak{f}_m(t) = (t - \zeta_s)(U_m(t) + U_{m-1}(t)), \quad (3.10)$$

and

$$q_{2m+1}(x, s) = a^m [\mathfrak{g}_m]_{u,v}, \quad \mathfrak{g}_m(t) = \left((x-s)t - \frac{x(1+a^2) + 2as}{2a} \right) U_m(t). \quad (3.11)$$

The recurrence and generating identities are polynomial and hence global. The real-pair formulas apply when the two symmetric relations hold, with the displayed derivative as the collision value at $u = v$. When $\Delta_{\text{fold}} < 0$, the polynomial identities avoid choosing a complex square-root branch.

3.2.1. Derivation of the folded formulas

Pair the coordinates in the order $(1, n), (2, n-1), \dots$. In this order the symmetric matrix $(xI_n - A_n)J_n - sI_n$ is block tridiagonal. Its interior diagonal and off-diagonal blocks are

$$D_{\text{int}} = \begin{bmatrix} -s & x \\ x & -s \end{bmatrix}, \quad E_{\text{off}} = \begin{bmatrix} 0 & -1 \\ -a & 0 \end{bmatrix},$$

with lower block E_{off}^T . For even order the terminal block is

$$D_{\text{term}}^e = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix};$$

for odd order the terminal block is the scalar $x - s$, coupled to the preceding block by the column $(-1, -a)^T$. Since

$$(xI_n - A_n - sJ_n) = ((xI_n - A_n)J_n - sI_n)J_n,$$

the determinant of this block matrix, multiplied by $\det J_n$, is $q_n(x, s)$. Set $\mathcal{Q}_m^e = q_{2m}(x, s)$ and $\mathcal{Q}_m^o = q_{2m+1}(x, s)$. We derive a recurrence by keeping track of five minors. For $\tau \in \{e, o\}$, let K_m^τ denote the corresponding matrix

$$(xI_n - A_n - sJ_n)J_n = (xI_n - A_n)J_n - sI_n, \quad n = 2m \text{ or } 2m + 1,$$

after simultaneously permuting its rows and columns into the paired order. Thus K_m^e has order $2m$, K_m^o has order $2m + 1$, and

$$K_1^e = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix}, \quad K_1^o = \begin{bmatrix} -s & x & -1 \\ x & -s & -a \\ -1 & -a & x-s \end{bmatrix}.$$

For $m \geq 1$, increasing m by one adds one interior block at the exposed end:

$$K_{m+1}^\tau = \begin{bmatrix} D_{\text{int}} & \mathcal{E}_m^\tau \\ (\mathcal{E}_m^\tau)^T & K_m^\tau \end{bmatrix}, \quad \mathcal{E}_m^\tau = \begin{bmatrix} E_{\text{off}} & 0 \end{bmatrix}, \quad (3.12)$$

where the zero block has the width required by K_m^τ .

For a matrix whose first two coordinates are exposed, write $K(\widehat{I} \mid \widehat{J})$ for the matrix obtained by deleting rows I and columns J . Suppressing the parity superscript, set

$$\begin{aligned} \mathfrak{d}_m &= \det K_m, & \mathfrak{p}_m &= \det K_m(\widehat{1} \mid \widehat{1}), & \mathfrak{q}_m &= \det K_m(\widehat{2} \mid \widehat{2}), \\ \mathfrak{c}_m &= \det K_m(\widehat{1} \mid \widehat{2}), & \mathfrak{r}_m &= \det K_m(\widehat{1}, \widehat{2} \mid \widehat{1}, \widehat{2}). \end{aligned}$$

The off-principal minor obtained by deleting row 2 and column 1 has the same value $\mathfrak{c}_m$, because K_m is symmetric.

The one-step transfer follows from a Schur complement. On the dense set where both K_m and the tail obtained by deleting its first two coordinates are invertible, let

$$F = \begin{bmatrix} u & w \\ w & v \end{bmatrix}$$

be its Schur complement in K_m , and let δ be the determinant of that tail. Then

$$(\mathfrak{d}_m, \mathfrak{p}_m, \mathfrak{q}_m, \mathfrak{c}_m, \mathfrak{r}_m) = \delta(uv - w^2, v, u, w, 1).$$

Since

$$E_{\text{off}} \text{adj}(F) E_{\text{off}}^T = \begin{bmatrix} u & -aw \\ -aw & a^2v \end{bmatrix},$$

taking the Schur complement of K_m in (3.12) gives

$$\begin{aligned} \mathfrak{d}_{m+1} &= (s^2 - x^2)\mathfrak{d}_m + s(a^2\mathfrak{p}_m + \mathfrak{q}_m) - 2ax\mathfrak{c}_m + a^2\mathfrak{r}_m, \\ \mathfrak{p}_{m+1} &= -s\mathfrak{d}_m - a^2\mathfrak{p}_m, \\ \mathfrak{q}_{m+1} &= -s\mathfrak{d}_m - \mathfrak{q}_m, \\ \mathfrak{c}_{m+1} &= x\mathfrak{d}_m + a\mathfrak{c}_m, \\ \mathfrak{r}_{m+1} &= \mathfrak{d}_m. \end{aligned} \quad (3.13)$$

Both sides are polynomial in x and s . For each fixed $0 < a < 1$, the identities hold on a dense nonsingular set, so the polynomial identity theorem extends them to all x, s , including singular matrix cases.

The five quantities in (3.13) satisfy one linear invariant. Define

$$\ell_m = (1-a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1-a)x\mathfrak{c}_m - a(1-a)\mathfrak{r}_m.$$

Direct substitution into (3.13) gives

$$\ell_{m+1} = -a\ell_m.$$

For the two terminal blocks, direct evaluation gives

$$\begin{aligned} & (\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_e \\ &= ((1+s)(a+s) - x^2, -a-s, -1-s, x, 1), \\ & (\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_o \\ &= (x(1+a^2) + 2as - C_0(x-s), \\ & \quad s^2 - sx - a^2, s^2 - sx - 1, x^2 - sx - a, x-s). \end{aligned}$$

Thus $\ell_1 = 0$ in both parity classes, and consequently

$$(1-a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1-a)x\mathfrak{c}_m - a(1-a)\mathfrak{r}_m = 0 \quad (m \geq 1). \quad (3.14)$$

Because

$$\det J_{2m} = \det J_{2m+1} = (-1)^m,$$

the first coordinate of the following signed state is the required determinant:

$$\boldsymbol{\eta}_m^\tau = (-1)^m \begin{bmatrix} \mathfrak{d}_m^\tau & \mathfrak{p}_m^\tau & \mathfrak{q}_m^\tau & \mathfrak{c}_m^\tau \end{bmatrix}^T, \quad (\boldsymbol{\eta}_m^e)_1 = \mathcal{Q}_m^e, \quad (\boldsymbol{\eta}_m^o)_1 = \mathcal{Q}_m^o.$$

Eliminating $\mathfrak{r}_m$ with (3.14) reduces (3.13) to the four-dimensional transfer

$$\boldsymbol{\eta}_{m+1}^\tau = \mathsf{T}_{\text{fold}} \boldsymbol{\eta}_m^\tau, \quad \mathsf{T}_{\text{fold}} = \begin{bmatrix} x^2 - s^2 - a & \frac{a^3 s}{1-a} & -\frac{s}{1-a} & ax \\ s & a^2 & 0 & 0 \\ s & 0 & 1 & 0 \\ -x & 0 & 0 & -a \end{bmatrix}. \quad (3.15)$$

Set

$$\boldsymbol{\eta}_0^e = \begin{bmatrix} 1 & a^{-1} & 1 & 0 \end{bmatrix}^T, \quad \boldsymbol{\eta}_0^o = \begin{bmatrix} x-s & 1 & 1 & -1 \end{bmatrix}^T. \quad (3.16)$$

These values incorporate the terminal conditions at $m = 0$. Direct multiplication in (3.15) reproduces the signed terminal minors above. Hence $\boldsymbol{\eta}_m^\tau = \mathsf{T}_{\text{fold}}^m \boldsymbol{\eta}_0^\tau$ for every $m \geq 0$.

A direct determinant calculation gives

$$\begin{aligned} \det(\lambda I_4 - \mathsf{T}_{\text{fold}}) &= \lambda^4 - C_0 \lambda^3 + \mathfrak{b} \lambda^2 - a^2 C_0 \lambda + a^4, \\ \mathfrak{b} &= (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3. \end{aligned} \quad (3.17)$$

Apply Cayley–Hamilton with the characteristic polynomial (3.17) to T_{fold} , multiply by $\mathsf{T}_{\text{fold}}^{m-4} \boldsymbol{\eta}_0^\tau$, and extract the first coordinate. For both $Z_m = \mathcal{Q}_m^e$ and $Z_m = \mathcal{Q}_m^o$, this yields

$$Z_m - C_0 Z_{m-1} + \mathfrak{b} Z_{m-2} - a^2 C_0 Z_{m-3} + a^4 Z_{m-4} = 0, \quad m \geq 4. \quad (3.18)$$

Applying T_{fold} three times to (3.16) gives the initial residuals for the recurrence (3.18):

$$\begin{aligned} \text{Res}_0 &= Z_0, & \text{Res}_1 &= Z_1 - C_0 Z_0, & \text{Res}_2 &= Z_2 - C_0 Z_1 + \mathfrak{b} Z_0, \\ \text{Res}_3 &= Z_3 - C_0 Z_2 + \mathfrak{b} Z_1 - a^2 C_0 Z_0, \end{aligned}$$

namely

	Res ₀	Res ₁	Res ₂	Res ₃
$Z_m = \mathcal{Q}_m^e$	1	$a(1 - 2\zeta_s)$	$a^2(1 - 2\zeta_s)$	a^3
$Z_m = \mathcal{Q}_m^o$	$x - s$	$-[x(1 + a^2) + 2as]$	$a^2(x - s)$	0

Thus, as a formal-power-series identity,

$$\Pi(z) \sum_{m \geq 0} Z_m z^m = \text{Res}_0 + \text{Res}_1 z + \text{Res}_2 z^2 + \text{Res}_3 z^3.$$

For the even sequence the polynomial on the right factors as $(1 + az)(1 - 2a\zeta_s z + a^2 z^2)$; for the odd sequence it is $(x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2$. These factorizations identify the boundary-residual polynomials. Since $\Pi(0) = 1$, division by Π in $\mathbb{R}[[z]]$ gives the formal-power-series identities

$$\sum_{m \geq 0} \mathcal{Q}_m^e z^m = (1 + az)(1 - 2a\zeta_s z + a^2 z^2) \Pi(z)^{-1}, \quad (3.19)$$

and

$$\sum_{m \geq 0} \mathcal{Q}_m^o z^m = ((x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2) \Pi(z)^{-1}. \quad (3.20)$$

The preceding transfer calculation gives two rational generating functions. Coefficient extraction converts them to the divided-difference formulas used in the half-angle chord parametrization. Fix $0 < a < 1$ and set $U_{-1} = 0$. The Chebyshev generating function gives

$$\sum_{m \geq 0} (u - \zeta_s)(U_m(u) + U_{m-1}(u)) \omega_{\text{gf}}^m = \frac{(u - \zeta_s)(1 + \omega_{\text{gf}})}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2},$$

and, with $B_s^{\text{fold}} = [x(1 + a^2) + 2as]/(2a)$,

$$\sum_{m \geq 0} ((x - s)u - B_s^{\text{fold}}) U_m(u) \omega_{\text{gf}}^m = \frac{(x - s)u - B_s^{\text{fold}}}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2}.$$

Let $u, v \in \mathbb{R}$ satisfy the two symmetric relations in (3.9). Taking the corresponding divided difference of the Chebyshev generating identities and putting $\omega_{\text{gf}} = az$ recovers (3.19) and (3.20). Coefficient comparison gives (3.10)–(3.11), including the derivative value when $u = v$. The underlying polynomial identities remain valid when $\Delta_{\text{fold}} < 0$, without choosing a square-root branch.

3.3. Half-angle chord parametrization

Put $h = -\frac{1}{2} \log a > 0$, so that $a = e^{-2h}$, and set

$$\mathfrak{s}_h = \sinh h, \quad \mathfrak{c}_h = \cosh h.$$

For $K \geq 2$, define

$$\begin{aligned} \Phi_K(\theta) &= \sqrt{\mathfrak{s}_h^2 + \sin^2 \theta} \frac{|\sin(K\theta)|}{\sin \theta}, & 0 \leq \theta \leq \pi, \\ \Psi_K(\eta) &= \sqrt{\mathfrak{s}_h^2 - \sinh^2 \eta} \frac{\sinh(K\eta)}{\sinh \eta}, & 0 \leq \eta \leq h. \end{aligned} \quad (3.21)$$

The quotients in (3.21) are interpreted continuously at $\theta = 0, \pi$ and $\eta = 0$. The following half-angle reduction also fixes the sign convention. Put

$$y = \cos \theta, \quad z = \cosh \eta, \quad p_\theta = \sqrt{c_h^2 - y^2}, \quad p_\eta = \sqrt{c_h^2 - z^2},$$

and let $L_0 = 2r/c_h$. If $x = L_0 y z$, $s = L_0 p_\theta p_\eta > 0$, and $\tau = \zeta s$ with $\zeta \in \{-1, 1\}$, then direct expansion gives

$$\begin{aligned} (1 + a + x)^2 - s^2 &= 4a(z + y)^2, & (1 + a - x)^2 - s^2 &= 4a(z - y)^2, \\ u = 2z^2 - 1, & v = 2y^2 - 1, & \zeta_\tau &= 2c_h^2 - 1 + 2\zeta p_\theta p_\eta. \end{aligned} \quad (3.22)$$

Substitution of (3.22) shows that the real pair (u, v) satisfies the two symmetric relations in (3.9). For $K = 2m + 1$, use

$$U_m(2t^2 - 1) + U_{m-1}(2t^2 - 1) = U_{K-1}(t);$$

for $K = 2m + 2$, use

$$U_m(2t^2 - 1) = \frac{U_{K-1}(t)}{2t}.$$

Substitution of this pair in (3.10) and (3.11), respectively, reduces either equation $q_{K-1}(x, \tau) = 0$ to

$$p_\theta U_{K-1}(y) = \zeta p_\eta U_{K-1}(z). \quad (3.23)$$

In the even-fold case the intermediate factorization is

$$(p_\eta + \zeta p_\theta)(p_\eta U_{K-1}(z) - \zeta p_\theta U_{K-1}(y)) = 0.$$

The first factor could vanish only if $\zeta = -1$ and $p_\eta = p_\theta$, which would imply $y^2 = z^2$; the distinct-variable condition therefore leaves precisely (3.23). In the odd-fold case the reduction also contains the factor $p_\eta y + \zeta p_\theta z$. On a distinct-variable sheet this factor is nonzero: if it vanished, squaring would give

$$(c_h^2 - z^2)y^2 = (c_h^2 - y^2)z^2,$$

and hence $y^2 = z^2$. Thus the factor is nonzero whenever the folded variables are distinct. On the outer lobe $U_{K-1}(z) > 0$, whereas, for $j\pi/K < \theta < (j+1)\pi/K$,

$$\operatorname{sgn}(U_{K-1}(\cos \theta)) = \operatorname{sgn}(\sin(K\theta)) = (-1)^j.$$

Moreover, $p_\theta |U_{K-1}(y)| = \Phi_K(\theta)$ and $p_\eta U_{K-1}(z) = \Psi_K(\eta)$. Thus (3.23) is equivalent to $\zeta = (-1)^j$ and

$$\Phi_K(\theta) = \Psi_K(\eta), \quad (3.24)$$

with

$$s = \mathcal{S}_{\text{hyp}}(\theta, \eta) := \frac{2r}{c_h} \sqrt{(s_h^2 + \sin^2 \theta)(s_h^2 - \sinh^2 \eta)}, \quad x = \frac{2r}{c_h} \cos \theta \cosh \eta. \quad (3.25)$$

Consequently $q_{K-1}(x, (-1)^j s) = 0$. Conversely, for distinct real folded variables, their sum and product determine

$$x^2 = L_0^2 y^2 z^2, \quad s^2 = L_0^2 (c_h^2 - y^2)(c_h^2 - z^2).$$

Thus, for distinct real variables and a fixed sign of x , the two chord relations are equivalent to the corresponding signed equation. The continuous divided-difference convention extends the determinant identities to a folded-variable collision; the selected middle branch is shown below to be collision-free. Here $K - 1$ is the matrix size. If the exterior argument has crossed through 1, write it as $z = \cos \phi$,

$0 \leq \phi < \pi/K$. Then the chord equation is $\Phi_K(\theta) = \Phi_K(\phi)$, and the second factor in the formula for s is $s_h^2 + \sin^2 \phi$. We denote the resulting value by

$$\mathcal{S}_{\text{ell}}(\theta, \phi) = \frac{2r}{c_h} \sqrt{(s_h^2 + \sin^2 \theta)(s_h^2 + \sin^2 \phi)}.$$

3.4. Outer-lobe geometry and chord classification

To identify the chords that realize the middle branch of Lemma 3.2, set

$$\mathcal{W}_K(u) = \sqrt{c_h^2 - u^2} |U_{K-1}(u)|, \quad |u| \leq c_h. \quad (3.26)$$

At $u = \pm c_h$, (3.26) denotes the continuous endpoint value 0. Then

$$\Phi_K(\theta) = \mathcal{W}_K(\cos \theta), \quad \Psi_K(\eta) = \mathcal{W}_K(\cosh \eta).$$

If $u_{\ell,K}^{(0)} = \cos \frac{\ell\pi}{K}$, then, away from the zeros,

$$\frac{d}{du} \log \mathcal{W}_K(u) = \sum_{\ell=1}^{K-1} \frac{1}{u - u_{\ell,K}^{(0)}} - \frac{u}{c_h^2 - u^2},$$

and its derivative is

$$-\sum_{\ell=1}^{K-1} \frac{1}{(u - u_{\ell,K}^{(0)})^2} - \frac{c_h^2 + u^2}{(c_h^2 - u^2)^2} < 0. \quad (3.27)$$

The strict negativity in (3.27) makes the logarithmic derivative strictly decreasing. At each finite nodal endpoint it tends to the appropriate signed infinity, and it tends to $-\infty$ as $u \uparrow c_h$. Thus there is exactly one lobe maximum in every nodal interval, including the outer interval $\mathcal{J}_K^{\text{out}} = (\cos \frac{\pi}{K}, c_h)$. Let u_K^{max} be the unique maximizer on $\mathcal{J}_K^{\text{out}}$. For every $0 \leq \ell < \mathcal{W}_K(u_K^{\text{max}})$, there is a unique solution

$$z_K(\ell) \in (u_K^{\text{max}}, c_h], \quad \mathcal{W}_K(z_K(\ell)) = \ell;$$

we call it the *endpoint-side outer solution*. If $z_K(\ell) \geq 1$, write it as $\cosh \eta$, $0 \leq \eta \leq h$; if it is at most 1, write it as $\cos \phi$, $0 \leq \phi < \pi/K$. These are two coordinates on the same outer lobe, joined at $z = 1$.

For a positive-half gap and $\theta \in (j\pi/K, (j+1)\pi/K)$, $\theta \leq \pi/2$, a *j-chord* is the pair

$$y = \cos \theta, \quad z = z_K(\mathcal{W}_K(y)), \quad \mathcal{W}_K(y) = \mathcal{W}_K(z),$$

together with the coordinate-free values

$$x = L_0 y z, \quad s = L_0 \sqrt{(c_h^2 - y^2)(c_h^2 - z^2)};$$

its signed root is $(-1)^j s$. The term endpoint-side always refers to the part of the outer lobe adjacent to c_h , independently of whether it is described by the hyperbolic or elliptic coordinate.

Reflection reduces the branch classification to a positive-half gap, so $j \geq 1$ and $\theta \leq \pi/2$. Put $\phi_{\text{rem}} = \theta - j\pi/K \in (0, \pi/K)$. Since $|\sin(K\phi_{\text{rem}})| = |\sin(K\theta)|$ and

$$\mathcal{A}_h(t) = \frac{\sqrt{s_h^2 + \sin^2 t}}{\sin t}$$

is strictly decreasing on $(0, \pi/2]$,

$$\Phi_K(\theta) < \Phi_K(\phi_{\text{rem}}) \leq \max_{u \in \mathcal{J}_K^{\text{out}}} \mathcal{W}_K(u). \quad (3.28)$$

Thus the endpoint-side outer solution exists for every inner parameter.

At $x = \lambda_{K-1,k} = 2r \cos \frac{k\pi}{K}$, $s = 0$, the two folded variables are $\cos \frac{2k\pi}{K}$ and $\cosh(2h)$, or, in half variables, $\cos \frac{k\pi}{K}$ and c_h . Moreover, zero is a simple eigenvalue of $B_{K-1}(\lambda_{K-1,k})$: the matrix $\lambda_{K-1,k}I_{K-1} - A_{K-1}$ has rank $K-2$, and J_{K-1} is invertible. The implicit-function theorem therefore gives, near either gap endpoint, a real continuation of the middle signed root and real half variables $y = \cos \theta$ and z .

3.5. Exhaustion of the selected chord sheet

Lemma 3.3 (Endpoint-side chord-sheet exhaustion). *Let $K \geq 3$ and let $j \geq 1$ be an integer. Under the conventions of Equations (3.24) and (3.25), if*

$$\frac{(j+1)\pi}{K} \leq \frac{\pi}{2},$$

then every point of the ordered middle branch in the corresponding open positive gap has a selected endpoint-side j -chord representation with $j\pi/K < \theta < (j+1)\pi/K$, and every θ in this interval gives a point of that branch. The signed root is $(-1)^j s$, its magnitude is $s_{\min}(xI_{K-1} - A_{K-1})$, and the two folded variables are distinct. This includes a gap adjacent to zero. Separately, if K is odd and $j = (K-1)/2$, the same bidirectional statement holds from the positive spectral endpoint to the center of the single central gap, with $j\pi/K < \theta \leq \pi/2$.

Proof. Continue the ordered middle root $s_*(x)$ through the open gap. Write $\Delta_*(x) = \Delta_{\text{fold}}(x, s_*(x))$ for its folded discriminant, let $y(x) \leq z(x)$ be the reconstructed nonnegative half variables on the real sheet issuing from the endpoint germ, and set $\theta_{\text{ch}}(x) = \arccos y(x)$. Let $\mathcal{C}$ be the set of gap points for which

$$\Delta_*(x) > 0, \quad \frac{j\pi}{K} < \theta_{\text{ch}}(x) < \frac{(j+1)\pi}{K}, \quad u_K^{\max} < z(x) < c_h.$$

On $\mathcal{C}$, the reversible folded determinant identities give $\mathcal{W}_K(y(x)) = \mathcal{W}_K(z(x))$; uniqueness on the endpoint-side outer lobe then gives

$$z(x) = z_K(\mathcal{W}_K(y(x))).$$

Thus $\mathcal{C}$ is precisely the selected endpoint-side chord sheet. At the upper gap endpoint, $z = c_h$ and $\theta_{\text{ch}} = j\pi/K$. There $z = c_h > 1 \geq y$, so $\Delta_* > 0$ persists immediately inside the gap. Moreover, Lemma 3.2 gives $\text{sgn } s_* = (-1)^j$; the signed chord equation therefore places θ_{ch} in $(j\pi/K, (j+1)\pi/K)$. Since $|s_*| > 0$, the channel identity gives $z < c_h$, while continuity keeps $z > u_K^{\max}$. Hence $\mathcal{C}$ is nonempty. The ordered eigenvalues of the continuous real symmetric pencil $B_{K-1}(x)$ are continuous in x , including at an eigenvalue crossing, so the continuation passes through a multiple least singular value. The folded pair $u = 2z^2 - 1$, $v = 2y^2 - 1$ and its reconstructed half variables are continuous, while the sheet inequalities are strict. Thus $\mathcal{C}$ is relatively open in the gap.

It is also relatively closed. At a relative closure point the chord identity and all sheet inequalities persist weakly. The possible closure cases are as follows. (1) If $\Delta_*(x) = 0$, then the reconstructed half variables coincide, which is impossible because

$$z \geq u_K^{\max} > \cos \frac{\pi}{K} \geq \cos \frac{j\pi}{K} \geq y.$$

(2) At an interior point of a positive gap, including one adjacent to zero, both half variables retain their signs: $z > 0$, while $y = 0$ would force $x = 0$, outside the open positive half. (3) Neither the outer variable can reach c_h nor the inner variable a nodal endpoint: in either case the common level is zero, the endpoint-side solution is $z = c_h$, and (3.25) gives $s = 0$, impossible inside a spectral gap. (4) The outer variable remains in its endpoint-side component, since its first exit point would be $u_K^{\max}$, contradicting (3.28). Passage through $z = 1$ changes only the coordinate representation. Hence every

relative closure point remains in $\mathcal{C}$. Since the open gap is an interval, $\mathcal{C}$ is the entire gap. This proves the first exhaustion direction: every actual gap point lies on the selected endpoint-side chord sheet.

For the converse direction, the reconstructed angle θ_{ch} extends continuously to the closed spectral gap. The exact folded variables at the two endpoints give

$$\theta_{\text{ch}}(\lambda_{K-1,j+1}) = \frac{(j+1)\pi}{K}, \quad \theta_{\text{ch}}(\lambda_{K-1,j}) = \frac{j\pi}{K}.$$

For every $\theta_0 \in (j\pi/K, (j+1)\pi/K)$, the intermediate value theorem therefore supplies an interior gap point x with $\theta_{\text{ch}}(x) = \theta_0$. The preceding classification and uniqueness of the endpoint-side outer solution then give

$$y(x) = \cos \theta_0, \quad z(x) = z_K(\mathcal{W}_K(\cos \theta_0)),$$

so the corresponding abscissa and signed root are exactly those of the prescribed j -chord. Thus every chord parameter produces an actual point of the ordered middle branch. By Lemma 3.2, its positive magnitude is $s_{\min}(xI_{K-1} - A_{K-1})$.

When K is odd, the same argument applies to the positive half of the single central gap, continuing from the positive spectral endpoint to $x = 0$. Because $z > 0$, the identity $x = (2r/c_h) \cos \theta z$ forces $\theta = \pi/2$ at $x = 0$. Thus every parameter on that half occurs, and reflection covers the negative half. $\square$

3.6. Noncentral gap comparison

Lemma 3.4 (Noncentral gap comparison). *Let $j \geq 1$ and $N \geq 2(j+1)$ be integers, and put $M = N+1$. For every $x \in G_{M-1,j}^\circ$, there is $\hat{x} \in G_{N-1,j}^\circ$ such that*

$$g_{N-1}(\hat{x}) > g_{M-1}(x).$$

Proof. Fix $x \in G_{M-1,j}^\circ$. By Lemma 3.3, its middle branch has a hyperbolic or elliptic chord representation with

$$\theta \in \left(\frac{j\pi}{M}, \frac{(j+1)\pi}{M} \right).$$

Set $\kappa = N/M$ and $\theta_0 = \theta/\kappa$. The hypotheses place θ_0 in the corresponding N -nodal interval, with every trigonometric argument below in $(0, \pi/2]$.

First suppose $\Phi_M(\theta) = \Psi_M(\eta)$. Concavity of sine gives

$$\frac{\Phi_N(\theta_0)}{\Phi_M(\theta)} = \frac{\mathcal{A}_h(\theta_0)}{\mathcal{A}_h(\theta)} > \kappa. \quad (3.29)$$

Indeed, $\theta = \kappa\theta_0$ and strict concavity gives $\sin \theta > \kappa \sin \theta_0$; hence

$$\mathcal{A}_h(\theta_0)^2 - \kappa^2 \mathcal{A}_h(\theta)^2 = (1 - \kappa^2) + \mathfrak{s}_h^2 \left(\frac{1}{\sin^2 \theta_0} - \frac{\kappa^2}{\sin^2 \theta} \right) > 0.$$

On the other hand,

$$\frac{\Psi_N(\eta)}{\Psi_M(\eta)} = \frac{\sinh(N\eta)}{\sinh(M\eta)} \leq \kappa, \quad (3.30)$$

with strict inequality for $\eta > 0$, by convexity of $\sinh$ on the positive half-axis. Together, (3.29) and (3.30) imply $\Phi_N(\theta_0) > \Psi_N(\eta)$. Since Φ_N vanishes at the right endpoint $(j+1)\pi/N$, its rightmost solution $\hat{\theta}$ of

$$\Phi_N(\hat{\theta}) = \Psi_N(\eta)$$

exists by the intermediate value theorem and satisfies $\hat{\theta} > \theta_0 > \theta$. Strict log-concavity of $\mathcal{W}_N$ makes it the unique solution on the descending half of that lobe. The same exterior variable is endpoint-side for the N -problem as well. Indeed, for $u = \cosh \eta > 1$,

$$\frac{d}{du} \log \mathcal{W}_M(u) - \frac{d}{du} \log \mathcal{W}_N(u) = \frac{M \coth(M\eta) - N \coth(N\eta)}{\sinh \eta} > 0.$$

The numerator is positive because, for $\eta > 0$,

$$\frac{d}{dt} (t \coth(t\eta)) = \frac{\sinh(t\eta) \cosh(t\eta) - t\eta}{\sinh^2(t\eta)} > 0.$$

The endpoint-side branch is characterized by a nonpositive logarithmic derivative. The displayed difference therefore makes the N -derivative strictly negative. The case $\eta = 0$ follows by continuity. The chord classification above says that the constructed value $\mathcal{S}_{\text{hyp}}(\hat{\theta}, \eta)$ is the least singular value at its constructed abscissa. Formula (3.25) and the strict increase of sine on the interval now give

$$\mathcal{S}_{\text{hyp}}(\hat{\theta}, \eta) > \mathcal{S}_{\text{hyp}}(\theta, \eta). \quad (3.31)$$

Suppose next that the exterior endpoint is elliptic, so that $\Phi_M(\theta) = \Phi_M(\phi)$, $0 < \phi < \pi/M$. (The case $\phi = 0$ is the limiting hyperbolic case already covered.) Put

$$\theta_0 = \theta/\kappa, \quad \phi_0 = \phi/\kappa.$$

The ratio

$$\mathcal{A}_{N,M}^\Phi(t) = \frac{\Phi_N(t/\kappa)}{\Phi_M(t)} = \frac{\mathcal{A}_h(t/\kappa)}{\mathcal{A}_h(t)}$$

is strictly increasing while $0 < t < t/\kappa \leq \pi/2$. Indeed,

$$\frac{d}{dt} \log \mathcal{A}_{N,M}^\Phi(t) = \mathfrak{d}_h(t) - \kappa^{-1} \mathfrak{d}_h(t/\kappa) > 0, \quad \mathfrak{d}_h(t) = \frac{s_h^2 \cot t}{s_h^2 + \sin^2 t},$$

because

$$t \mathfrak{d}_h(t) = (t \cot t) \frac{s_h^2}{s_h^2 + \sin^2 t}$$

is strictly decreasing: both positive factors decrease on $(0, \pi/2)$, and the second decreases strictly. Since $\theta > \pi/M > \phi$, it follows that $\Phi_N(\theta_0) > \Phi_N(\phi_0)$. The rightmost solution of

$$\Phi_N(\hat{\theta}) = \Phi_N(\phi_0)$$

therefore has $\hat{\theta} > \theta_0$. Its exterior variable is endpoint-side. Indeed, if $\Lambda_K(t) = (\log \Phi_K)'(t)$, then on the endpoint-side part of the first outer lobe $\Lambda_M(\phi) \geq 0$, and differentiation of $\Phi_N(t/\kappa) = \mathcal{A}_{N,M}^\Phi(t) \Phi_M(t)$ gives

$$\kappa^{-1} \Lambda_N(t/\kappa) = \Lambda_M(t) + \frac{(\mathcal{A}_{N,M}^\Phi)'(t)}{\mathcal{A}_{N,M}^\Phi(t)} > 0.$$

Because $\Lambda_K(\phi) = -\sin \phi (\log \mathcal{W}_K)'(\cos \phi)$, positivity of $\Lambda_N(\phi_0)$ is exactly the endpoint-side condition for the N -outer solution. The one-maximum property now fixes the label, and the chord classification says again that $\mathcal{S}_{\text{ell}}(\hat{\theta}, \phi_0)$ is the least singular value at the constructed abscissa. Here $\hat{\theta} > \theta_0 > \theta$ and $\phi_0 > \phi$, with all four angles below $\pi/2$. Thus both sine factors in the elliptic version of (3.25) have increased, at least one strictly, and hence

$$\mathcal{S}_{\text{ell}}(\hat{\theta}, \phi_0) > \mathcal{S}_{\text{ell}}(\theta, \phi). \quad (3.32)$$

Equations (3.31) and (3.32), together with the coordinate formula for x , place the constructed abscissa $\hat{x}$ in $G_{N-1,j}^\circ$. Chord exhaustion identifies the original and constructed heights with $g_{M-1}(x)$ and $g_{N-1}(\hat{x})$, respectively, proving the claim. $\square$

3.7. Central gap comparison

For $m \geq 1$, define the even-order central height by

$$\mathfrak{h}_m^e = s_{\min}(aI_m + V_m). \quad (3.33)$$

By reflection, the two gaps adjacent to the zero eigenvalue of A_{2m+1} have the same height; denote it by

$$\mathfrak{h}_m^o = \max_{0 \leq x \leq 2r \sin \frac{\pi}{2m+2}} s_{\min}(xI_{2m+1} - A_{2m+1}). \quad (3.34)$$

The central heights defined in (3.33) and (3.34) satisfy the following strict alternation.

Lemma 3.5 (Central gap comparison). *For every $m \geq 1$,*

$$\mathfrak{h}_m^e > \mathfrak{h}_m^o > \mathfrak{h}_{m+1}^e. \quad (3.35)$$

Proof. Put

$$\zeta = \frac{a + a^{-1}}{2}.$$

For every $j \geq 1$, there is a unique $\rho_j > \cos \frac{\pi}{j+1}$ such that

$$aU_j(\rho_j) = U_{j-1}(\rho_j). \quad (3.36)$$

Indeed, the Christoffel–Darboux identity gives, above the largest zero of U_j ,

$$\frac{d}{dt} \frac{U_{j-1}(t)}{U_j(t)} = -\frac{2 \sum_{k=0}^{j-1} U_k(t)^2}{U_j(t)^2} < 0.$$

The ratio decreases from $+\infty$ to 0, proving existence and uniqueness. A three-term determinant recurrence gives

$$\det\left((aI_j + V_j)^*(aI_j + V_j) - (1 + a^2 - 2at)I_j\right) = a^j \left(U_j(t) - a^{-1}U_{j-1}(t)\right). \quad (3.37)$$

Thus the largest root ρ_j gives the least eigenvalue. Together with

$$U_j(t)^2 + U_{j-1}(t)^2 - 2tU_j(t)U_{j-1}(t) = 1,$$

equations (3.36) and (3.37) give

$$(\mathfrak{h}_j^e)^2 = 1 + a^2 - 2a\rho_j = \frac{1}{U_j(\rho_j)^2}, \quad \zeta - \rho_j = \frac{1}{2U_{j-1}(\rho_j)U_j(\rho_j)}. \quad (3.38)$$

We first identify $\mathfrak{h}_m^e$ as the central-gap height of A_{2m} . Let

$$P_{\text{oe}} = [\mathbf{e}_1, \mathbf{e}_3, \dots, \mathbf{e}_{2m-1}, \mathbf{e}_2, \mathbf{e}_4, \dots, \mathbf{e}_{2m}]$$

be the permutation matrix that places odd coordinates first. Then

$$P_{\text{oe}}^T A_{2m} P_{\text{oe}} = \begin{bmatrix} 0 & I_m + aV_m^T \\ aI_m + V_m & 0 \end{bmatrix}.$$

Moreover,

$$(I_m + aV_m^T)^*(I_m + aV_m^T) - J_m(aI_m + V_m)^*(aI_m + V_m)J_m = (1 - a^2)\mathbf{e}_m\mathbf{e}_m^T \geq 0.$$

The singular values of the block off-diagonal matrix are those of its two off-diagonal blocks. The Gram inequality shows that $s_{\min}(I_m + aV_m^T) \geq s_{\min}(aI_m + V_m) = \mathfrak{h}_m^e$, and hence $s_{\min}(A_{2m}) = \mathfrak{h}_m^e$. To maximize over the central gap, put $K = 2m + 1$ and use the half variables

$$y = \cos \theta, \quad z = \cosh \eta \quad \text{or} \quad z = \cos \phi.$$

On the positive half of the central middle branch, the preceding chord classification gives

$$\mathcal{W}_K(y) = \mathcal{W}_K(z), \quad x = \frac{2r}{\mathfrak{c}_h}yz, \quad s^2 = \frac{4a}{\mathfrak{c}_h^2}(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2), \quad (3.39)$$

where $0 \leq y < \cos \frac{m\pi}{K}$ is in the central lobe and z is the endpoint-side outer solution. The function $\mathcal{W}_K$ is even, and strict log-concavity on the central lobe gives

$$\mathcal{W}_K(y) < \mathcal{W}_K(0) = \mathfrak{c}_h \quad (y > 0).$$

Let

$$z_0 = \sqrt{\frac{1 + \rho_m}{2}}.$$

The half-angle identity and (3.36) give

$$U_{2m}(z_0) = U_m(\rho_m) + U_{m-1}(\rho_m) = (1 + a)U_m(\rho_m).$$

Together with (3.38), this yields

$$\mathcal{W}_K(z_0) = \mathfrak{c}_h, \quad 4a(\mathfrak{c}_h^2 - z_0^2) = 2a(\zeta - \rho_m) = (\mathfrak{h}_m^e)^2. \quad (3.40)$$

At $x = 0$, Lemma 3.2 and chord exhaustion identify the nearest root with $\mathfrak{h}_m^e$, so z_0 is the endpoint-side outer solution of level $\mathfrak{c}_h$. If $x \neq 0$, the level in (3.39) is strictly below $\mathfrak{c}_h$, and its endpoint-side solution satisfies $z > z_0$. Equations (3.39)–(3.40) then give

$$s^2 < 4a(\mathfrak{c}_h^2 - z_0^2) = (\mathfrak{h}_m^e)^2.$$

Reflection gives the same strict bound on the other half of the gap. Therefore the central-gap height is $\mathfrak{h}_m^e$, uniquely attained at $x = 0$.

Set

$$\omega_m^e = \cos \frac{\pi}{m+1}, \quad \mathcal{V}_m(t) = \sqrt{(\zeta - t)(1 + t)} U_m(t).$$

For the central specialization, take $K = 2m + 2$ and set

$$t_{\text{in}} = 2y^2 - 1 = \cos(2\theta), \quad t_{\text{out}} = 2z^2 - 1.$$

On the positive central gap, $-1 < t_{\text{in}} < -\omega_m^e$, while the endpoint-side outer variable satisfies $\omega_m^e < t_{\text{out}} < \zeta$, where $\zeta = 2\mathfrak{c}_h^2 - 1$. The identity $U_{2m+1}(t) = 2tU_m(2t^2 - 1)$ gives

$$\begin{aligned} \mathcal{W}_{2m+2}(y)^2 &= (\zeta - t_{\text{in}})(1 + t_{\text{in}})U_m(t_{\text{in}})^2, \\ \mathcal{W}_{2m+2}(z)^2 &= (\zeta - t_{\text{out}})(1 + t_{\text{out}})U_m(t_{\text{out}})^2. \end{aligned}$$

Because $\zeta + 1 = 2c_h^2$, the formulas $x = (2r/c_h)yz$ and $s^2 = (4a/c_h^2)(c_h^2 - y^2)(c_h^2 - z^2)$ become the following chord parametrization:

$$\begin{aligned} -1 < t_{\text{in}} < -\omega_m^e, \quad \omega_m^e < t_{\text{out}} < \zeta, \quad \mathcal{V}_m(t_{\text{out}}) = |\mathcal{V}_m(t_{\text{in}})|, \\ x^2 = \frac{2a(1+t_{\text{out}})(1+t_{\text{in}})}{\zeta+1}, \quad s^2 = \frac{2a(\zeta-t_{\text{out}})(\zeta-t_{\text{in}})}{\zeta+1}, \end{aligned} \quad (3.41)$$

where t_{out} is the descending, or largest, solution. To derive (3.41) from (3.24), use

$$s_h^2 + \sin^2 \theta = \frac{\zeta - t_{\text{in}}}{2}, \quad s_h^2 - \sinh^2 \eta = \frac{\zeta - t_{\text{out}}}{2}.$$

Continuation through $t_{\text{out}} = 1$ uses the elliptic coordinate $z = \cos \phi$.

Let $\rho = \rho_m$. The function $\log \mathcal{V}_m$ is strictly concave on (ω_m^e, ζ) . If $(\log \mathcal{V}_m)'(\rho) \geq 0$, then ρ lies at or before the unique lobe maximum. The descending root lies at or after that maximum, so $t_{\text{out}} \geq \rho$. Now suppose $(\log \mathcal{V}_m)'(\rho) < 0$. Then $\rho > a$: if $\rho \leq a < 1$, then $\rho < 1$, hence $\rho^2 - 1 \neq 0$, and the following quotient is well defined:

$$\frac{U'_m(\rho)}{U_m(\rho)} = \frac{m\rho - (m+1)a}{\rho^2 - 1}$$

would give

$$\begin{aligned} (\log \mathcal{V}_m)'(\rho) &= \frac{(2m+2)a + 1 - (2m+1)\rho}{2(1-\rho^2)} - \frac{1}{2(\zeta-\rho)} \\ &\geq \frac{1}{2(1-\rho)} - \frac{1}{2(\zeta-\rho)} \geq 0, \end{aligned}$$

which contradicts the assumed negative derivative. Hence $\rho > a$. Writing $t_{\text{in}} = -\cos \phi_c$, $0 < \phi_c < \pi/(m+1)$, gives

$$\begin{aligned} |\mathcal{V}_m(t_{\text{in}})|^2 &= \frac{(\zeta + \cos \phi_c) \sin^2((m+1)\phi_c)}{1 + \cos \phi_c} \\ &< \frac{\zeta + \omega_m^e}{1 + \omega_m^e} < \frac{1+a}{2a} < \frac{1+\rho}{2a} = \mathcal{V}_m(\rho)^2. \end{aligned}$$

Now ρ is on the descending side and $|\mathcal{V}_m(t_{\text{in}})| < \mathcal{V}_m(\rho)$, so the unique descending root again satisfies $t_{\text{out}} > \rho$. In either case, $t_{\text{out}} \geq \rho$, and $t_{\text{in}} > -1$ makes the following inequality strict:

$$s^2 < 2a(\zeta - \rho) = (\mathfrak{h}_m^e)^2.$$

This holds along the whole central sheet, so $\mathfrak{h}_m^o < \mathfrak{h}_m^e$. Thus the odd-order central height lies strictly below the preceding even-order central height. We next compare it with the following even-order height.

Lower comparison $\mathfrak{h}_m^o > \mathfrak{h}_{m+1}^e$: scalar estimate

Put $L = m+1 \geq 2$, and abbreviate

$$\begin{aligned} \rho = \rho_L, \quad u_- = U_{L-1}(\rho), \quad u_+ = U_L(\rho), \quad \omega_L = \cos \frac{\pi}{2L}, \quad \Delta_\omega = 1 - \omega_L, \\ \mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+, \quad \xi = \rho - \frac{\Delta_\omega}{\mathfrak{D}_L}. \end{aligned} \quad (3.42)$$

Thus $a = u_-/u_+$ and $\mathfrak{h}_L^e = 1/u_+$.

The required inequality is

$$(1 + \omega_L)(1 + \xi)^2 U_{L-1}(\xi)^2 < (\rho + \omega_L) \mathfrak{D}_L. \quad (3.43)$$

Put $b = u_+/u_- = 1/a$, and define

$$Q_b = b^2 - 2b\rho + 1, \quad S_b = b^2 + 2b\omega_L + 1.$$

The Chebyshev identity gives $u_-^2 Q_b = 1$, and

$$\mathfrak{D}_L = S_b/Q_b, \quad \Delta_\xi := \rho - \xi = \Delta_\omega Q_b/S_b. \quad (3.44)$$

Indeed, since $u_+ = bu_-$ and $u_-^2 = Q_b^{-1}$,

$$\mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+ = 1 + \frac{2b(\rho + \omega_L)}{Q_b} = \frac{S_b}{Q_b},$$

which proves the first identity in (3.44); the second follows from (3.42). The ratio U_L/U_{L-1} is strictly increasing above the largest zero of U_{L-1} , and equals 1 at $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Hence $\rho > \rho_{\text{ref}}$. Since $S_b - Q_b = 2b(\rho + \omega_L) > 0$,

$$\xi > \rho_{\text{ref}} - \Delta_\omega > 2\omega_L - 1 > 2\omega_L^2 - 1 = \cos \frac{\pi}{L}.$$

Therefore $f_L(x) = (1+x)U_{L-1}(x)$ is positive, strictly increasing, and strictly log-concave on $[\xi, \rho]$. In particular, $\mathfrak{d}_\rho = (\log f_L)'(\rho) > 0$, and

$$\mathfrak{d}_\rho = \begin{cases} \frac{L(b-\rho)-1}{\rho^2-1}, & \rho \neq 1, \\ \frac{2L^2+1}{6}, & \rho = 1 \end{cases}.$$

The endpoint value follows directly from $U_{L-1}(1) = L$ and $U'_{L-1}(1) = L(L^2-1)/3$. Moreover,

$$\log f_L(\rho) - \log f_L(\xi) > \Delta_\xi \mathfrak{d}_\rho, \quad f_L(\xi)^2 < f_L(\rho)^2 e^{-2\Delta_\xi \mathfrak{d}_\rho} < \frac{f_L(\rho)^2}{1 + 2\Delta_\xi \mathfrak{d}_\rho}.$$

Here the first strict inequality is the integral form of strict log-concavity. Since $f_L(\rho)^2 = (1+\rho)^2/Q_b$ and $\Delta_\xi = \Delta_\omega Q_b/S_b$, substitution into (3.43) reduces it to

$$\mathcal{N} = (\rho + \omega_L)(S_b + 2\Delta_\omega Q_b \mathfrak{d}_\rho) - (1 + \omega_L)(1 + \rho)^2 > 0. \quad (3.45)$$

All three cases use

$$\Delta_\omega = 1 - \cos \frac{\pi}{2L} < \frac{\pi^2}{8L^2} < \frac{5}{4L^2}, \quad \Delta_\omega L < 1. \quad (3.46)$$

Suppose first that $\rho > 1$. Put $u = b - \rho$ and $\Delta_+ = \rho^2 - 1$. Since $\mathfrak{d}_\rho > 0$, $u > 1/L$. Keeping ρ fixed, direct differentiation gives

$$\frac{\partial_b \mathcal{N}}{2(\rho + \omega_L)} = \rho + \omega_L + u - \Delta_\omega L + \frac{\Delta_\omega(3Lu^2 - 2u)}{\Delta_+}.$$

The right-hand side is strictly increasing for $u \geq 1/L$ and is positive at $u = 1/L$ by (3.46). Hence $\mathcal{N}$ exceeds its value at $b = \rho + 1/L$. With $s_0 = \rho + \omega_L$ and $\omega_+ = 1 + \omega_L$, that value is

$$\Upsilon(s_0) = s_0 \left[\left(s_0 + \frac{1}{L} \right)^2 + \omega_+ \Delta_\omega \right] - \omega_+(s_0 + \Delta_\omega)^2.$$

For $s_0 \geq \omega_+$,

$$\Upsilon''(s_0) = 6s_0 + 4/L - 2\omega_+ > 0, \quad \Upsilon'(\omega_+) = 2\omega_L\omega_+ + 4\omega_+/L + 1/L^2 > 0,$$

and

$$Y(\omega_+) = \omega_+(-2\Delta_\omega + 2\omega_+/L + 1/L^2) > 0.$$

Thus (3.45) holds when $\rho > 1$. If $\rho = 1$, then $b = (L + 1)/L$, $Q_b = L^{-2}$, and direct substitution, using the endpoint value of $\mathfrak{d}_\rho$, gives

$$\mathcal{N} = Y(1 + \omega_L) + \frac{2(1 + \omega_L)\Delta_\omega}{L^2} \frac{2L^2 + 1}{6} > 0.$$

The first term is positive by the preceding calculation, and every factor in the correction term is positive. For $\rho < 1$, set

$$v = 1 - \rho, \quad \Delta_b = b - 1, \quad u = \Delta_b + v, \quad \Delta_- = v(2 - v).$$

Because $b > 1$, we have $0 < v < u$. Moreover, $\rho^2 - 1 < 0$, so the positivity of $\mathfrak{d}_\rho$ in its quotient formula forces $L(b - \rho) - 1 < 0$. Since $u = b - \rho$,

$$0 < v < u < 1/L. \quad (3.47)$$

With ρ fixed, write $\mathcal{N}_\rho^<(\cdot)$ for $\mathcal{N}$ viewed as a function of $\Delta_b = b - 1$. A direct expansion of (3.45) gives

$$\mathcal{N}_\rho^<(\Delta_b) - \mathcal{N}_\rho^<(0) = \frac{\Delta_b(\rho + \omega_L)}{\Delta_-} q(u),$$

where

$$q(u) = \Delta_-[2(2 - \Delta_\omega) + u - v] + 2\Delta_\omega[u + v - L(u^2 + uv + v^2) - L\Delta_-].$$

The function q is a concave quadratic. Moreover,

$$\mathcal{N}_\rho^<(0) = \frac{v}{2 - v} [(2 - \Delta_\omega)(2 - v)^2 - 2\Delta_\omega^2 - 4\Delta_\omega L(2 - v - \Delta_\omega)].$$

The following elementary bounds hold; the second follows from $\rho > \rho_{\text{ref}}$ and $1 - \rho_{\text{ref}} < \pi^2/[2(2L + 1)^2] < 5/(2L + 1)^2$:

$$\Delta_\omega < \frac{5}{4L^2}, \quad v < \frac{5}{(2L + 1)^2} \leq \frac{1}{5}. \quad (3.48)$$

When $L = 2$, the non-strict bound equals $1/5$, while the preceding strict estimate still gives $v < 1/5$. These bounds in (3.48) show that the bracketed expression is positive, since it equals

$$(2 - v)[(2 - \Delta_\omega)(2 - v) - 4\Delta_\omega L] + 2\Delta_\omega^2(2L - 1)$$

and, using $L \geq 2$,

$$(2 - \Delta_\omega)(2 - v) > \frac{27}{16} \frac{9}{5} = \frac{243}{80} > \frac{5}{2} \geq 4\Delta_\omega L.$$

At the endpoints of (3.47),

$$q(v) = v[(2 - v)(4 - 2\Delta_\omega) + 4\Delta_\omega - 4\Delta_\omega L(1 + v)] > 0,$$

$$q(1/L) = v[(2 - v)(4 - 2\Delta_\omega + 1/L - v) - 4\Delta_\omega L] > 0.$$

At the two endpoints, the positive leading terms exceed

$$\frac{9}{5} \left(4 - \frac{5}{8}\right) = \frac{243}{40}, \quad \frac{9}{5} \left(4 - \frac{5}{8} - \frac{1}{5}\right) = \frac{1143}{200},$$

while the negative terms have magnitudes less than $(5/L)(6/5) \leq 3$ and $5/L \leq 5/2$, respectively. Concavity therefore gives $q(u) > 0$, proving (3.45), and hence (3.43), in all three cases.

A point above the next even-order height

Since $a > 0$, $\Delta_\omega > 0$, $\rho + \omega_L > 0$, and $\zeta + \omega_L > 0$, let $x_* > 0$ be the positive square root determined by

$$x_*^2 = \frac{2a\Delta_\omega(\rho + \omega_L)}{\zeta + \omega_L}, \quad \tau_* = \mathfrak{h}_L^e. \quad (3.49)$$

At $x = x_*$ and $s = \pm\tau_*$, where $\tau_* = \mathfrak{h}_L^e$, the real pair $u = \zeta$, $v = -\omega_L$ satisfies the two symmetric relations in (3.9). Indeed, its sum and product are $\zeta - \omega_L$ and $-\omega_L\zeta$, and the required identities follow from

$$\mathfrak{D}_L = \frac{\zeta + \omega_L}{\zeta - \rho}, \quad \zeta = \frac{\rho(\zeta + 1) + \zeta(\omega_L - 1)}{\zeta + \omega_L}.$$

Positivity of $\zeta + \omega_L$ selects the distinct-variable branch of the divided difference in (3.11). Write the folded numerator as

$$\mathfrak{n}_s(t) = -[x(\zeta - t) + s(1 + t)]U_{L-1}(t).$$

Put

$$X_\zeta = x_*(\zeta - \zeta), \quad Y_\zeta = \tau_*(1 + \zeta), \quad X_\omega = x_*(\zeta + \omega_L), \quad Y_\omega = \tau_*\Delta_\omega,$$

and set $U_\zeta = U_{L-1}(\zeta)$, $U_{-\omega} = U_{L-1}(-\omega_L)$. The identities

$$\mathfrak{D}_L = 2a(\zeta + \omega_L)u_+^2, \quad \mathfrak{D}_L(\rho + \omega_L)(\zeta - \zeta) = (\zeta + \omega_L)(1 + \zeta)$$

give $Y_\omega Y_\zeta = X_\omega X_\zeta$. The identity $u_-^2 Q_b = 1$ gives $Q_b > 0$, and $S_b - Q_b = 2b(\rho + \omega_L) > 0$, so $\mathfrak{D}_L = S_b/Q_b > 1$. Moreover, $L \geq 2$ gives $\omega_L > 1/2$, while $\rho > \rho_{\text{ref}} > \omega_L$; hence

$$\rho + \omega_L > 2\omega_L > 1 > 1 - \omega_L > 0.$$

Consequently,

$$\frac{X_\omega^2}{Y_\omega^2} = \frac{\mathfrak{D}_L(\rho + \omega_L)}{1 - \omega_L} > 1,$$

so, with $\mathfrak{r}_* = X_\zeta/Y_\zeta = Y_\omega/X_\omega \in (0, 1)$, the common factor $a^{L-1}/(\zeta + \omega_L)$ in (3.11) is positive, and the two numerator differences are

$$\begin{aligned} \mathfrak{n}_{\tau_*}(\zeta) - \mathfrak{n}_{\tau_*}(-\omega_L) &= (1 + \mathfrak{r}_*)(X_\omega U_{-\omega} - Y_\zeta U_\zeta), \\ \mathfrak{n}_{-\tau_*}(\zeta) - \mathfrak{n}_{-\tau_*}(-\omega_L) &= (1 - \mathfrak{r}_*)(X_\omega U_{-\omega} + Y_\zeta U_\zeta). \end{aligned}$$

Their product is positive by (3.43), because

$$U_{-\omega}^2 = \frac{1}{1 - \omega_L^2}, \quad X_\omega^2 U_{-\omega}^2 = \frac{\tau_*^2 \mathfrak{D}_L(\rho + \omega_L)}{1 + \omega_L} > Y_\zeta^2 U_\zeta^2.$$

Thus

$$q_{2L-1}(x_*, \tau_*)q_{2L-1}(x_*, -\tau_*) > 0. \quad (3.50)$$

At $x = 0$, the second-smallest singular value of A_{2L-1} is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

Indeed, after separating odd and even coordinates, the two rectangular off-diagonal blocks have Gram matrices, taken in their smaller dimensions, equal to the $(L-1)$ -dimensional tridiagonal Toeplitz matrix with diagonal $1 + a^2$ and off-diagonal a . Their singular values therefore have squares

$$1 + a^2 + 2a \cos \frac{k\pi}{L}, \quad 1 \leq k \leq L-1,$$

and each occurs twice, while the remaining singular value is zero. We claim that

$$x_* + \tau_* < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}. \quad (3.51)$$

Indeed, recall $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Since $\rho > \rho_{\text{ref}}$ and U_{L-1} is strictly increasing above its largest zero,

$$u_- = U_{L-1}(\rho) > U_{L-1}(\rho_{\text{ref}}) =: u_{-, \text{ref}}, \quad u_{-, \text{ref}}^{-2} = 2(1 - \rho_{\text{ref}}).$$

The identities $\tau_* = a/u_-$ and $\tau_*^2 = 2a(\zeta - \rho)$ imply $\zeta - \rho < a(1 - \rho_{\text{ref}}) < 1 - \rho_{\text{ref}}$. Also

$$\frac{x_*^2}{2a} = \Delta_\omega \frac{\rho + \omega_L}{\zeta + \omega_L} < \Delta_\omega.$$

Put

$$\theta_L = \frac{\pi}{4L}, \quad \Delta_L = \frac{\theta_L}{2L+1}.$$

Then $\Delta_\omega = 2 \sin^2 \theta_L$ and $1 - \rho_{\text{ref}} = 2 \sin^2(\theta_L - \Delta_L)$. Since all quantities are positive, the preceding quotient bound for x_*^2 gives

$$x_* < 2\sqrt{a} \sin \theta_L.$$

Likewise,

$$\tau_*^2 = 2a(\zeta - \rho) < 2a^2(1 - \rho_{\text{ref}}) = 4a^2 \sin^2(\theta_L - \Delta_L),$$

and hence

$$\tau_* < 2a \sin(\theta_L - \Delta_L) < 2\sqrt{a} \sin(\theta_L - \Delta_L),$$

where the last inequality uses $0 < a < 1$, hence $a < \sqrt{a}$. It suffices to prove

$$\sin \theta_L + \sin(\theta_L - \Delta_L) < \sin(2\theta_L). \quad (3.52)$$

The estimates

$$2 \sin \theta_L - \sin(2\theta_L) = 2 \sin \theta_L (1 - \cos \theta_L) < \theta_L^3$$

and

$$\sin \theta_L - \sin(\theta_L - \Delta_L) = \int_{\theta_L - \Delta_L}^{\theta_L} \cos u \, du > \Delta_L \cos \theta_L$$

reduce (3.52) to $\Delta_L \cos \theta_L > \theta_L^3$. Substituting $\Delta_L = \theta_L / (2L+1)$ and $\theta_L = \pi / (4L)$ gives the equivalent condition $2 \cos \theta_L > \theta_L(\pi + 2\theta_L)$. For $0 < \theta_L \leq \pi/8$, using $\pi^2 < 10$,

$$2 \cos \theta_L > 2 - \theta_L^2 \geq \frac{59}{32} > \frac{25}{16} \geq \theta_L(\pi + 2\theta_L).$$

Consequently $x_* + \tau_* < 2\sqrt{a} \sin \frac{\pi}{2L}$. Finally,

$$(s_{L,2}^{(0)})^2 - 4a \sin^2 \frac{\pi}{2L} = (1-a)^2 \geq 0,$$

which proves (3.51).

Let $\mu_1, \dots, \mu_{2L-1}$ be the eigenvalues of the real symmetric matrix

$$B_{2L-1}(x_*) = (x_* I_{2L-1} - A_{2L-1}) J_{2L-1}.$$

Right multiplication by J_{2L-1} preserves singular values, so the numbers $|\mu_i|$ are precisely the singular values of $x_* I_{2L-1} - A_{2L-1}$. Moreover,

$$q_{2L-1}(x_*, \tau_*) q_{2L-1}(x_*, -\tau_*) = \prod_{i=1}^{2L-1} (\mu_i^2 - \tau_*^2).$$

Positivity of the product in (3.50) implies that the number of indices with $|\mu_i| < \tau_*$ is even. On the other hand, Weyl's singular-value perturbation inequality and (3.51) give, for the second-smallest singular value s_{next} ,

$$s_{\text{next}}(x_* I_{2L-1} - A_{2L-1}) \geq s_{L,2}^{(0)} - x_* > \tau_*,$$

so at most one index can satisfy $|\mu_i| < \tau_*$. Parity therefore gives $|\mu_i| \geq \tau_*$ for every i , and strict positivity of the product rules out equality. Hence $s_{\min}(x_* I_{2L-1} - A_{2L-1}) > h_L^e$. Since $0 < x_* < 2r \sin \frac{\pi}{2L}$, the point lies strictly in the positive central gap, and $h_{L-1}^o > h_L^e$. Together with the upper comparison $h_m^e > h_m^o$, this proves (3.35). $\square$

3.8. Assembly of the global comparison

Proof of Lemma 3.1. Reflection and the central interlacing chain now assemble the local comparisons into the global barrier inequality. Put $N = n + 1$ and $M = N + 1$, so the larger and smaller matrices have orders $M - 1$ and $N - 1$, respectively. If $\Sigma = \text{diag}(1, -1, 1, -1, \dots)$, then $\Sigma A_n \Sigma = -A_n$, so reflection reduces every noncentral comparison to a positive gap. For each positive noncentral gap $G_{M-1,j}$, choose $x_j \in G_{M-1,j}^\circ$ attaining $\gamma_{M-1,j}$. Then $j \geq 1$ and $(j+1)\pi/N \leq \pi/2$, so Lemma 3.4 produces a point $\hat{x}_j \in G_{N-1,j}^\circ$, and chord exhaustion gives

$$\gamma_{N-1,j} \geq g_{N-1}(\hat{x}_j) > g_{M-1}(x_j) = \gamma_{M-1,j}.$$

Reflection gives the same inequality on the corresponding reflected gap.

By Lemma 3.5, if $M = 2\ell_* + 1$, the remaining single central height of the larger, even-order matrix is $h_{\ell_*}^e$, whereas each central height of the smaller odd-order matrix is $h_{\ell_*-1}^o > h_{\ell_*}^e$. If $M = 2\ell_* + 2$, the two central heights of the larger odd-order matrix equal $h_{\ell_*}^o < h_{\ell_*}^e$, where $h_{\ell_*}^e$ is the single central height of the smaller even-order matrix. Thus every height of the larger problem is strictly smaller than at least one height of the smaller problem. By (3.1), choose j_* with $\gamma_{M-1,j_*} = \max_j \gamma_{M-1,j}$. Then

$$\gamma_{M-1} = \gamma_{M-1,j_*} < \max_k \gamma_{N-1,k} = \gamma_{N-1}.$$

Since $M - 1 = n + 1$ and $N - 1 = n$, this proves $\gamma_{n+1} < \gamma_n$. $\square$

4. Quantitative bounds and the transition scale

The connectedness criterion (1.6) and the strict comparison of Lemma 3.1 reduce the remaining problem to estimating γ_n . Put $N = n + 1$, and define

$$\mathcal{U}_n(a) = r^n \csc \frac{\pi}{n+1}, \tag{4.1}$$

$$\mathcal{L}_n(a) = \left(1 + r^2 - 2r \cos \frac{\pi}{n+1}\right) \frac{r^n(1-r^2)}{(1-r^{2(n+1)}) \sin \frac{3\pi}{2(n+1)}}. \tag{4.2}$$

Proposition 4.1 (Optimal-order gap bounds). *For every $n \geq 2$,*

$$\mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a). \tag{4.3}$$

In particular, both bounds have order $(n+1)a^{n/2}$, with constants depending only on a .

Proof. We prove the two inequalities in (4.3) separately. For $x = 2r \cos \theta \in I_n^{\text{sp}}$, set

$$v = (v_j)_{j=1}^n, \quad v_j = r^{j-1} \sin(j\theta), \quad 1 \leq j \leq n.$$

The residual $(xI_n - A_n)v$ is supported in its last coordinate, where it equals $r^n \sin(N\theta)$. Thus

$$g_n(2r \cos \theta) \leq \frac{r^n |\sin(N\theta)|}{\left(\sum_{j=1}^n r^{2j-2} \sin^2(j\theta)\right)^{1/2}} \leq r^n \frac{|\sin(N\theta)|}{\sin \theta}.$$

Since $\pi/N \leq \theta \leq (N-1)\pi/N$, this proves the upper bound.

For the lower bound, put $p_\theta = re^{i\theta}$, $q_\theta = re^{-i\theta}$, so that $p_\theta + q_\theta = x$ and $p_\theta q_\theta = a$. For $\mathbf{u} = (u_j)_{j=1}^n \in \mathbb{C}^n$, put $u_0 = u_{n+1} = 0$, and define maps from $\mathbb{C}^n$ to $\mathbb{C}^{n+1}$ by

$$(\mathbf{C}_\theta \mathbf{u})_j = p_\theta u_j - u_{j+1}, \quad (\mathbf{D}_\theta \mathbf{u})_j = u_j - \overline{q_\theta} u_{j+1}, \quad 0 \leq j \leq n.$$

Direct multiplication, using $p_\theta + q_\theta = x$ and $p_\theta q_\theta = a$, gives

$$xI_n - A_n = \mathbf{D}_\theta^* \mathbf{C}_\theta. \quad (4.4)$$

The two Gram matrices associated with the factorization (4.4) are diagonally unitarily similar to the tridiagonal Toeplitz matrix with diagonal $1 + r^2$ and off-diagonal $-r$. Its eigenvalues are $1 + r^2 - 2r \cos \frac{k\pi}{N}$, $1 \leq k \leq n$. Consequently, for every $\mathbf{u} \in \mathbb{C}^n$,

$$\begin{aligned} \|\mathbf{C}_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2, \\ \|\mathbf{D}_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2. \end{aligned} \quad (4.5)$$

Put

$$\mathcal{S}_N = \sum_{j=0}^n r^{2j} = \frac{1 - r^{2N}}{1 - r^2}.$$

Unit normal vectors to the two hyperplane ranges are

$$\mathbf{u}_C = \mathcal{S}_N^{-1/2} (\overline{p_\theta}^n, \overline{p_\theta}^{n-1}, \dots, 1)^T, \quad \mathbf{u}_D = \mathcal{S}_N^{-1/2} (1, q_\theta, \dots, q_\theta^n)^T.$$

In the polar decompositions

$$\mathbf{C}_\theta = Q_C (\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}, \quad \mathbf{D}_\theta = Q_D (\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2},$$

the matrices Q_C, Q_D are isometries onto these ranges. Two hyperplanes in $\mathbb{C}^{n+1}$ intersect in dimension at least $n-1$, so the cross-Gram matrix $Q_D^* Q_C$ has $n-1$ singular values equal to 1; its final singular value is $|\langle \mathbf{u}_D, \mathbf{u}_C \rangle|$, the cosine of the nontrivial principal angle. The geometric sum gives

$$\begin{aligned} |\langle \mathbf{u}_D, \mathbf{u}_C \rangle| &= \frac{r^n}{\mathcal{S}_N} \left| \sum_{j=0}^{N-1} e^{i(2j-(N-1))\theta} \right| \\ &= \frac{r^n (1 - r^2)}{1 - r^{2N}} \left| \frac{\sin(N\theta)}{\sin \theta} \right| =: \chi_N(\theta). \end{aligned} \quad (4.6)$$

Finally,

$$xI_n - A_n = (\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2} Q_D^* Q_C (\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}.$$

Using $s_{\min}(XYZ) \geq s_{\min}(X)s_{\min}(Y)s_{\min}(Z)$ and (4.5), with χ_N as defined in (4.6), gives

$$g_n(2r \cos \theta) \geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \chi_N(\theta).$$

Putting $\theta = 3\pi/(2N)$, which lies in the first spectral gap, proves the lower bound.

Finally,

$$\mathcal{U}_n(a) \leq \frac{1}{2}(n+1)r^n,$$

while

$$\mathcal{L}_n(a) \geq C_{\text{lb}}(a)(n+1)r^n, \quad C_{\text{lb}}(a) = \frac{2(1-r)^2(1-r^2)}{3\pi}. \quad (4.7)$$

The estimate (4.7) and the preceding upper estimate follow from $\sin \frac{\pi}{N} \geq 2/N$, $1+r^2-2r \cos \frac{\pi}{N} \geq (1-r)^2$, and $\sin \frac{3\pi}{2N} \leq 3\pi/(2N)$. This proves the asserted optimal order up to constants depending only on a . $\square$

The bounds also locate the first connected order:

$$1 + \max(\{1\} \cup \{n \geq 2 : \mathcal{L}_n(a) \geq \varepsilon\}) \leq N_c(a, \varepsilon), \quad (4.8)$$

$$N_c(a, \varepsilon) \leq \min\{n \geq 2 : \mathcal{U}_n(a) < \varepsilon\}.$$

Proof of Theorem 1.1. By Proposition 4.1, $\gamma_n \leq \mathcal{U}_n(a) \rightarrow 0$. Lemma 3.1 says that γ_n is strictly decreasing. Equation (1.6) therefore shows that, once the pseudospectrum is connected, it stays connected at every larger matrix order. Contractibility and the connectedness criterion were proved in Proposition 2.4.

If $\mathcal{L}_n(a) \geq \varepsilon$, then $\gamma_n \geq \varepsilon$, so the pseudospectrum at order n is disconnected. If $\mathcal{U}_n(a) < \varepsilon$, then $\gamma_n < \varepsilon$, so it is connected. Strict decrease of γ_n now gives (4.8).

For $t > 0$, define

$$\mathfrak{e}_n = (n+1)r^n, \quad \mathcal{T}_r(t) = \min\{N_0 \geq 2 : \mathfrak{e}_m < t \text{ for every } m \geq N_0\}.$$

This minimum exists because $\mathfrak{e}_m \rightarrow 0$. Because persistence gives

$$N_c(a, \varepsilon) = \min\{N \geq 2 : \gamma_m < \varepsilon \text{ for every } m \geq N\},$$

the bounds $C_{\text{lb}}(a)\mathfrak{e}_m \leq \gamma_m \leq \mathfrak{e}_m/2$ give

$$\mathcal{T}_r(\varepsilon/C_{\text{lb}}(a)) \leq N_c(a, \varepsilon) \leq \mathcal{T}_r(2\varepsilon). \quad (4.9)$$

Elementary inversion of the envelope threshold

Let

$$\beta = -\log r = \frac{1}{2} \log(1/a), \quad L(t) = \log(1/t), \quad S_r(t) = \frac{L(t) + \log L(t)}{\beta}.$$

For every fixed $D \in \mathbb{R}$, direct substitution gives

$$\beta(S_r(t) + D) - \log(S_r(t) + D + 1) - L(t) = \beta D + \log \beta + o(1) \quad (t \downarrow 0).$$

Indeed, $\beta S_r = L + \log L$ and $\log(S_r + D + 1) = \log L - \log \beta + o(1)$. Choose constants $D_- < -(\log \beta)/\beta < D_+$. For all sufficiently small t , the preceding residual is negative at $S_r(t) + D_-$ and positive at $S_r(t) + D_+$. Since $S_r(t) \rightarrow \infty$ and

$$\frac{d}{dx}((x+1)e^{-\beta x}) = e^{-\beta x}(1 - \beta(x+1)),$$

this function is decreasing on $[S_r(t) + D_-, \infty)$ for all sufficiently small t . The strict integer tail condition therefore gives

$$\lfloor S_r(t) + D_- \rfloor < \mathcal{T}_r(t) \leq \lceil S_r(t) + D_+ \rceil.$$

Thus $\mathcal{T}_r(t) = S_r(t) + O_r(1)$. Replacing t by a fixed positive multiple changes $S_r(t)$ by only $O_r(1)$; applying this to the two sides of (4.9) proves (1.8).

The threshold at order two

For $|x| \leq r$,

$$\det((xI_2 - A_2)^*(xI_2 - A_2) - a^2 I_2) = x^2(x^2 - 2a - 2a^2) \leq 0.$$

The matrix inside the determinant is Hermitian, and the nonpositive product of its two eigenvalues forces its least eigenvalue to be nonpositive. Thus $s_{\min}(xI_2 - A_2) \leq a$. At $x = 0$, the singular values of $-A_2$ are 1 and a , so equality is attained and $\gamma_2 = a$. Hence (1.6) gives $N_c(a, \varepsilon) = 2$ if and only if $\varepsilon > a$. This proves all assertions. $\square$

5. The boundary cases

The same matrices at $a = 0$ and $a = 1$ have simpler pseudospectra.

Proposition 5.1. *Let $n \geq 2$ and $\varepsilon > 0$. The pseudospectrum of $A_n(0)$ is an open disk centered at zero. For $a = 1$, every pseudospectral component is contractible, and the pseudospectrum is connected exactly when $\varepsilon > \eta_n$, where*

$$\eta_n = \begin{cases} 2 \sin \frac{\pi}{2(n+1)}, & n \text{ even}, \\ \sin \frac{\pi}{n+1}, & n \text{ odd}. \end{cases} \quad (5.1)$$

Moreover, $\eta_{n+1} < \eta_n$ for all $n \geq 2$, and $\eta_n = \pi/(n+1) + O(n^{-3})$.

Proof. At $a = 0$, the matrix is the nilpotent shift V_n . The proof of Lemma 2.3 applies unchanged, so every pseudospectral component contains its only eigenvalue, zero; the set is therefore connected. For $|q| = 1$, the unitary matrix $D(q) = \text{diag}(1, q, \dots, q^{n-1})$ satisfies $D(q)^* V_n D(q) = q V_n$. The pseudospectrum is consequently invariant under all rotations. Its set of moduli is a connected interval containing zero; boundedness and openness make it $[0, R)$ for some $R > 0$. Hence it is an open disk.

At $a = 1$, the matrix is real symmetric, and its pseudospectrum is the union of open disks of radius ε centered at $2 \cos(k\pi/(n+1))$, $1 \leq k \leq n$. It is connected precisely when twice the radius exceeds every adjacent eigenvalue gap. The identity

$$\cos \frac{k\pi}{n+1} - \cos \frac{(k+1)\pi}{n+1} = 2 \sin \frac{(2k+1)\pi}{2(n+1)} \sin \frac{\pi}{2(n+1)}$$

gives (5.1). Since contraction of the imaginary part decreases the distance to every eigenvalue, the retraction argument of Proposition 2.4 proves contractibility of each component.

For the transition from $n = 2m$ to $2m+1$, set $x = \pi/(4m+2)$ and $y = \pi/(2m+2)$. Then $y < 2x < \pi/2$, so $\sin y < \sin(2x) < 2 \sin x$. For the transition from $2m+1$ to $2m+2$, put $p = m+1 \geq 2$, $x = \pi/(2p)$, and $y = \pi/(4p+2)$. The inequalities $\sin x > x - x^3/6$, $\sin y < y$, and $24p^2 > \pi^2(2p+1)$ give $\sin x > 2 \sin y$. The last inequality holds because $24p^2/(2p+1)$ increases for $p \geq 2$ and equals $96/5 > \pi^2$ at $p = 2$. Finally, applying $0 \leq t - \sin t \leq t^3/6$ to each parity formula yields

$$0 \leq \frac{\pi}{n+1} - \eta_n \leq \frac{\pi^3}{6(n+1)^3},$$

which proves the stated asymptotic. $\square$

Thus the connectedness threshold is zero at $a = 0$, has order $(n+1)a^{n/2}$ for fixed $0 < a < 1$, and has order $(n+1)^{-1}$ at $a = 1$.

6. Numerical illustrations

Figures 1 and 2 illustrate two aspects of the matrix-order transition: coalescence of pseudospectral components and decay of the real-axis backward-error barrier.

At each sampled point, the least singular value was computed by a double-precision singular value decomposition. The contour plots use a 561-by-361 Cartesian grid. Each spectral gap was scanned at 121 equally spaced points before bounded scalar optimization was used to estimate γ_n . The displayed component counts were also evaluated on 1001 points per gap, with the 1-Lipschitz dependence of the least singular value controlling variation between adjacent sample points.

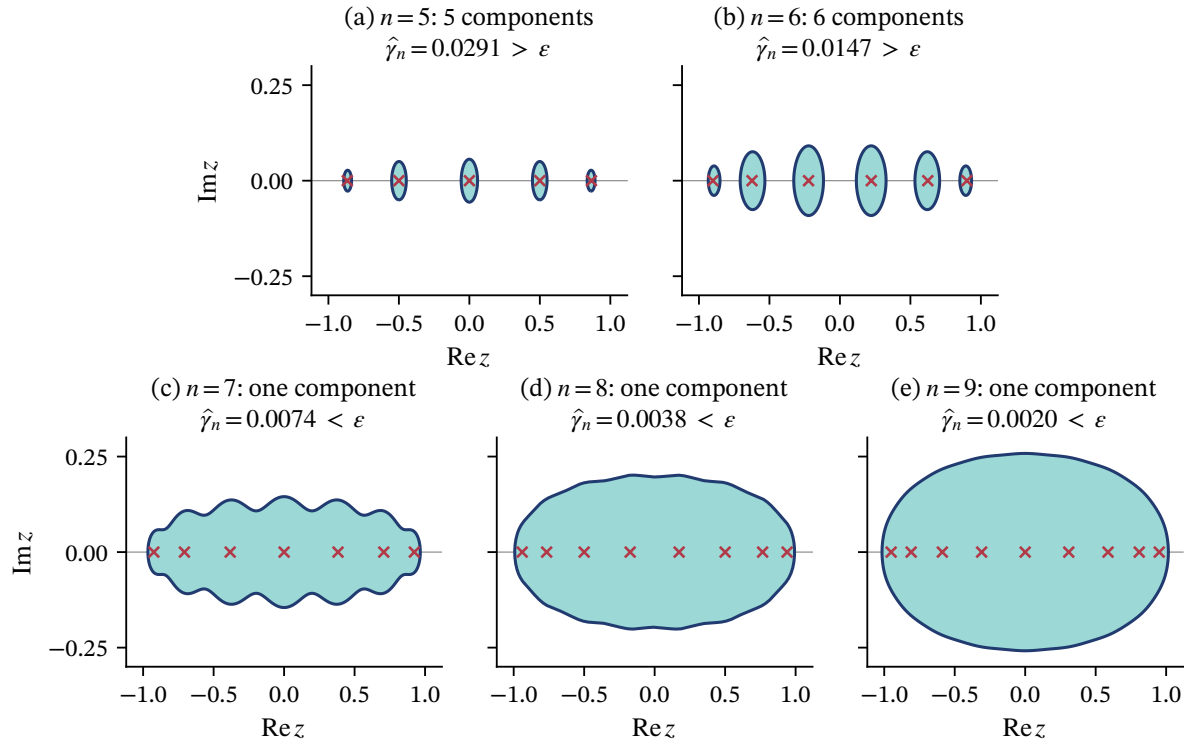


Figure 1. Numerically sampled open pseudospectra of $A_n(1/4)$ at $\epsilon = 10^{-2}$. Shading marks the sampled pseudospectrum and crosses mark the exact eigenvalues. The samples show component coalescence between $n = 6$ and $n = 7$.

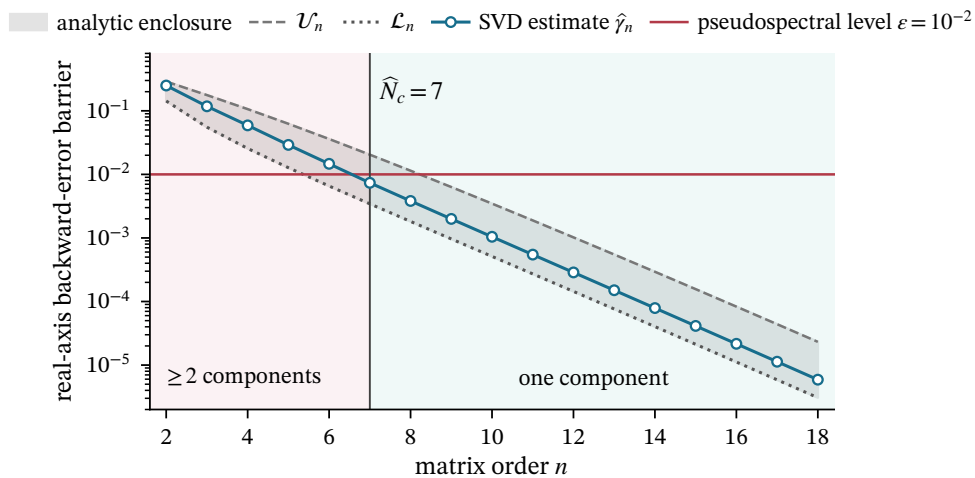


Figure 2. Real-axis backward-error barriers for $a = 1/4$. The gray band, bounded by the dotted lower curve $\mathcal{L}_n(a)$ and dashed upper curve $\mathcal{U}_n(a)$, is the rigorous enclosure of γ_n and establishes the order $(n+1)a^{n/2}$. Blue circles joined by a solid line are floating-point estimates; the red horizontal line is the fixed pseudospectral level $\epsilon = 10^{-2}$, and the black vertical line marks the numerical crossing $\hat{N}_c = 7$. The theorem gives the bracket $6 \leq N_c \leq 9$.

7. Concluding remarks

For $A_n(a)$, contraction toward the real axis reduces connectedness to the maximum of one real function over the spectral interval. The signed pencil then permits a strict comparison of these maxima at consecutive orders, despite the change in central-gap geometry. The bounds in Proposition 4.1 identify the polynomial factor multiplying the exponential decay and yield the second logarithmic term in the first connected order.

The asymptotic formula is for fixed $0 < a < 1$. A joint limit in which a approaches one as n increases would have to connect the exponential scale of Section 4 with the algebraic scale of Section 5.

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